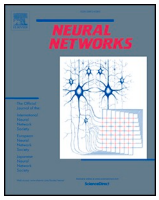
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Full Length Article



Synaptic pruning facilitates online Bayesian model selection

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ABSTRACT

Identifying appropriate structures for generative or world models is essential for both biological organisms and machines. This work shows that synaptic pruning facilitates efficient statistical structure learning. We extend previously established canonical neural networks to derive a synaptic pruning scheme that is formally equivalent to an online Bayesian model selection. The proposed scheme, termed Bayesian synaptic model pruning (BSyMP), utilizes connectivity parameters to switch between the presence (ON) and absence (OFF) of synaptic connections. Mathematical analyses reveal that these parameters converge to zero for uninformative connections, thus providing reliable and efficient model reduction. This enables the identification of a plausible structure for the environmental model, particularly when the environment is characterized by sparse likelihood and transition matrices. Through causal inference and rule learning simulations, we demonstrate that BSyMP achieves model reduction more efficiently than the conventional Bayesian model reduction scheme. These findings indicate that synaptic pruning could be a neuronal substrate underlying structure learning and generalizability in the brain.

1. Introduction

Characterizing brain intelligence is a significant challenge in the field of biology. Biological organisms recognize the external milieu by inferring underlying structures. Despite the importance of structure learning—the process of identifying an appropriate structure to represent the environment (Braun et al., 2010; Lynn & Bassett, 2020; Tenenbaum et al., 2011)—its neuronal mechanisms in the brain are not yet fully understood (Tervo et al., 2016). The structure of neural networks is shaped by the pruning of synaptic connections during development (Sakai, 2020), and several computational studies have explored the advantages of synaptic pruning for memory and learning (Chechik et al., 1998; Scholl et al., 2021; Thomas et al., 2015). However, analyzing the relationship between biologically grounded synaptic pruning and information-theoretically derived structure learning is still required.

The free-energy principle has been proposed in theoretical neuroscience to account for various brain functions, including perception, learning, and action, in terms of minimizing variational free energy (Friston, 2010; Friston et al., 2006). It models the brain as an agent that performs variational Bayesian inference of external states and parameters by minimizing the variational free energy. These inferences rest upon a generative or world model that describes how latent variables in the external milieu generate sensory inputs. Recent works have shown that the dynamics of canonical neural networks that have a certain biological plausibility can be conceptualized as variational Bayesian inference under implicit generative models characterized by network structures (Isomura et al., 2022; Isomura & Friston, 2020). This lends the explainability of Bayesian inference to investigate the functional properties of biologically plausible neural networks. Additionally, Bayesian inference can be employed in selecting model structures (Friston et al., 2017, 2019, 2024; Jafarian et al., 2019; Marković et al., 2024). However, the neural substrate of Bayesian model selection is yet to be established.

Herein, we show that biologically plausible synaptic pruning enables structure learning in a Bayes-optimal manner by extending the previously established equivalence between neural networks and Bayesian inference. We reveal that incorporating synaptic pruning into a canonical neural network is equivalent to online Bayesian model selection, a method we term Bayesian synaptic model pruning (BSyMP). BSyMP adopts a set of connectivity parameters that enables switching between the presence (ON) and absence (OFF) of the model parameters. These connectivity parameters, in addition to states and other parameters, are updated by minimizing the variational free energy that is equivalent to a cost function for canonical neural networks. Thereby, BSyMP optimizes

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model structures via variational Bayesian inference of the hyper- or connectivity parameters of the generative model. Mathematical analyses show that BSyMP is guaranteed to prune parameters unnecessary for propagating hidden state information, resulting in efficient model reduction.

We compared the performance of BSyMP with that of the previously proposed Bayesian model reduction (BMR). BMR reduces the number of parameters in a learned full model by minimizing the variational free energy in a post hoc manner (Friston et al., 2019). Owing to online and continuous (i.e., gradual) pruning, BSyMP outperformed BMR in numerical simulations of causal inference and rule learning. BSyMP could deliver successful results even when the full model and BMR failed. Finally, we discuss possible neuronal substrates that realize BSyMP-like synaptic pruning.

2. Methods

2.1. Equivalence between neural networks and Bayesian inference

The derivation of BSyMP is based on the equivalence between the dynamics of the neural networks and variational Bayesian inference established in the previous works (Isomura & Friston, 2020; Isomura et al., 2022). Here, we summarize the core concepts of these works prior to describing the generative model of BSyMP in subsequent subsections. In the previous works, a recurrent network of N_x neurons that receives N_o-dimensional binary observations (sensory inputs) was considered (Fig. 1, top left). The neural activity in the recurrent layer was formulated as follows:

$$\dot{x}(\tau) \propto -\mathrm{sig}^{-1}(x(\tau)) + (W_1 - W_0)o(\tau) + (K_1 - K_0)x(\tau - \Delta t) + h_1 - h_0, \tag{1}$$

where $x(\tau) := (x_1(\tau), \ldots, x_{N_x}(\tau))^{\mathrm{T}}$ denotes the firing rates of the neurons; $o(\tau) := (o_1(\tau), \ldots, o_{N_o}(\tau))^{\mathrm{T}}$ denotes binary observations; $\mathrm{sig}^{-1}(x(\tau))$ denotes the inverse function of the sigmoid function of $x(\tau)$ that models the leakage current; $W_1, W_0 \in \mathbb{R}^{N_x \times N_o}$ and $K_1,\ K_0 \in \mathbb{R}^{N_x \times N_x}$ are the feedforward and recurrent synaptic strength matrices, respectively; $\Delta t > 0$ is the delay in the recurrent input; h_1 and h_0 represent the adaptive firing thresholds that depend on the synaptic strengths. The variables with subscripts 1 and 0 separate the positive (excitatory) and negative (inhibitory) effects on neural activity. This neural activity equation can be derived from realistic neuron models, such as the Hodgkin-Huxley model and the FitzHugh-Nagumo model, through approximations (FitzHugh, 1961; Hodgkin & Huxley, 1952; Isomura et al., 2022; Nagumo et al., 1962). Here, the binary sensory inputs were considered in accordance with the equivalence to the partially observable Markov decision process (POMDP), which is widely used in the literature (Kurniawati, 2022) . The equivalence is explained later in this subsection. Such binary signals can be associated with ON and OFF of sensory receptor cells.

The cost function L for this network is a function that satisfies a requirement: the gradient descent on that yields the dynamics of the network. Therefore, it was derived by integrating Eq. (1) with respect to neural activity x and time τ:

$$L = \int_0^t \left\{ \begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix}^T \left\{ \ln \begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix} - \ln \begin{pmatrix} \widehat{W}_1 & \overline{\widehat{W}_1} \\ \widehat{W}_0 & \overline{\widehat{W}_0} \end{pmatrix} \begin{pmatrix} o(\tau) \\ \bar{o}(\tau) \end{pmatrix} \right.\right. \\ \left.\left. - \ln \begin{pmatrix} \widehat{K}_1 & \overline{\widehat{K}_1} \\ \widehat{K}_0 & \overline{\widehat{K}_0} \end{pmatrix} \begin{pmatrix} x(\tau - \Delta\tau) \\ \bar{x}(\tau - \Delta\tau) \end{pmatrix} - \begin{pmatrix} \phi_1 \\ \phi_0 \end{pmatrix} \right\} \right\} d\tau + \mathscr{O}(1), \tag{2}$$

where $\widehat{X} := \mathrm{sig}(X)$ denotes the sigmoid function and $\overline{X}$ is defined as $\overline{X} :=$

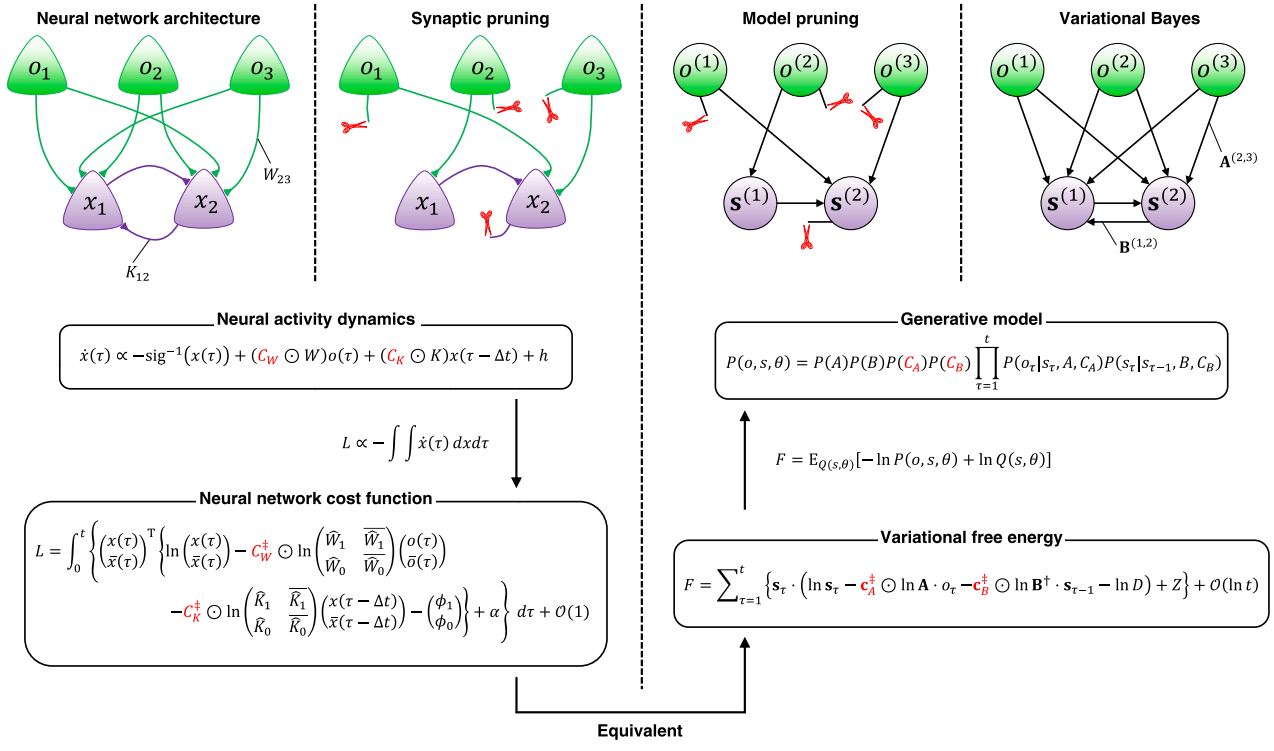


Fig. 1. Equivalence between synaptic pruning in neural networks and Bayesian pruning of connections in a generative model. Top left: A recurrent neural network receives sensory inputs. The occurrence of synaptic pruning is mediated by the connectivity parameters C_W and C_K. Middle left: The neural activity dynamics are given by Eq. (6). Bottom left: The neural network cost function L is derived by integrating Eq. (6) with respect to x and τ. The reconstructed L involves a free parameter α (i.e., integration constant). Bottom right: The variational free energy F equivalent to L is identified by extending the previously established equivalence between F and L. Middle right: The generative model $P(o,\ s,\ \theta)$ is reverse engineered from F. This enables the identification of the normalization factor Z that is involved in F. Top right: A graphical structure of the generative model is optimized through model pruning mediated by the connectivity parameters C_A and C_B. All variables are introduced in the main text and Table 1.The details of the notations and derivations are provided in Methods **2.1, 2.2, 2.3**, and Appendix A.

$1-X$ for any variable X. Moreover, $\phi_l := h_l - \ln\widehat{\overline{W_l}}\overrightarrow{1} - \ln\widehat{\overline{K_l}}\overrightarrow{1}$ $(l=1,0)$, where $\overrightarrow{1}$ is a vector of ones $\overrightarrow{1} := (1,\ldots,1)^{\mathrm{T}}$. The details of the derivations of Eq. (2) (i.e. integration of Eq. (1)) are provided in Appendix A.

In addition to the neural network, the environment was modeled as the POMDP. In this model, N_s binary hidden states $s_\tau := \left(s_\tau^{(1)},\ldots,s_\tau^{(N_s)}\right)^{\mathrm{T}}$ depend on the previous states $s_{\tau-1}$, following the state transition matrix B in the form of a categorical distribution (also called a generalized Bernoulli distribution), $P(s_\tau|s_{\tau-1},B)=\mathrm{Cat}(Bs_{\tau-1})$ (Bishop, 2006). The prior belief about initial states is described by the block vector D, $P(s_1)=\mathrm{Cat}(D)$. The hidden states generate N_o binary observations $o_\tau := \left(o_\tau^{(1)},\ldots,o_\tau^{(N_o)}\right)^{\mathrm{T}}$ through the likelihood mapping A, $P(o_\tau|s_\tau,A)=\mathrm{Cat}(As_\tau)$. The prior distributions of A and B follow the Dirichlet distributions $P(A)=\mathrm{Dir}(a)$, $P(B)=\mathrm{Dir}(b)$ (Bishop, 2006). Hence, the generative model is as follows:

$$P(o_{1:t},\ s_{1:t},A,B)=P(A)P(B)\prod_{\tau=1}^{t}P(o_\tau|s_\tau,A)P(s_\tau|s_{\tau-1},B). \tag{3}$$

The shape of the matrix A is in general $2N_o\times 2^{N_s}$ because the N_o binary observations are determined by the combination of the N_s binary states. However, here it was assumed that A and B were factorized into the products of 2×2 matrices $A^{(i,j)}$ and $B^{(i,j)}$. For instance, $A_{k\bullet}^{(i)}=A_{k\bullet}^{(i,1)}\otimes\cdots\otimes A_{k\bullet}^{(i,N_s)}$ denotes a row of the matrix A regarding i th observation. Under this assumption, A is considered as a set of $N_o\times N_s$ matrices (i.e., a block matrix), each block of which is 2×2 matrix $A^{(i,j)}$. A simplified notation for the factorial structure was adopted here; please refer to previous works (Isomura et al., 2022; Isomura & Friston, 2020) and Appendix B for the details of the generative model with the factorial structure.

Variational Bayesian inference is a process of optimizing the following approximate posterior belief:

$$Q(s_{1:t},A,B)=Q(A)Q(B)\prod_{\tau=1}^{t}Q(s_\tau). \tag{4}$$

The marginal posteriors about A, B, s_τ follow the Dirichlet distributions $Q(A)=\mathrm{Dir}(\mathbf{a})$, $Q(B)=\mathrm{Dir}(\mathbf{b})$ and categorical distributions $Q(s_\tau)=\mathrm{Cat}(\mathbf{s}_\tau)$ respectively. Variational free energy F under the above-mentioned generative model and posterior beliefs is given as follows:

$$F=\sum_{\tau=1}^{t}\left\{\mathbf{s}_\tau\cdot(\ln\mathbf{s}_\tau-\ln\mathbf{A}\cdot o_\tau-\ln\mathbf{B}^{\dagger}\cdot\mathbf{s}_{\tau-1}-\ln D)\right\}+\mathscr{O}(\ln t), \tag{5}$$

where $\ln\mathbf{A}:=\mathrm{E}_{Q(A)}[\ln A]$ denotes the posterior expectation of $\ln A$, and $\mathbf{B}^{\dagger}:=\mathbf{B}^{\mathrm{T}}\mathrm{diag}(D)^{-1}$ denotes the inverse transition matrix, in which $\ln\mathbf{B}s_{\tau-1}=\ln\mathbf{B}^{\dagger}\cdot\mathbf{s}_{\tau-1}+\ln D$ holds.

The previous works established a formal equivalence between Eq. (2) and (5) by showing one-to-one correspondences between their components, indicating that the neural network that minimizes the cost function L can be interpreted as an agent that performs variational Bayesian inference by minimizing the free energy F. Note that although the environmental states evolve in discrete time, the neural network activity exhibits the continuous time dynamics following the delay differential equation.

2.2. Derivation of generative model from neural network

Herein, we consider an extension of the conventional canonical neural networks (Isomura et al., 2022) by incorporating connectivity parameters (C_W and C_K) that express the presence (ON) or absence (OFF) of synaptic connections, which is formulated as follows:

$$\dot{x}(\tau)\propto -\mathrm{sig}^{-1}(x(\tau))+\{C_W\odot(W_1-W_0)\}o(\tau)+\{C_K\odot(K_1-K_0)\}x(\tau-\Delta t)+h_1-h_0, \tag{6}$$

where the symbol $\odot$ denotes the element-wise (Hadamard) product. The additional parameter matrices $C_W\in[0,1]^{N_x\times N_o}$ and $C_K\in[0,1]^{N_x\times N_x}$ express the probability of the presence of synaptic connections, providing a major difference from the previous work. As shown in Eq. (6), the signal from the corresponding synapse does not affect the neural activity when a certain element in C_W takes the value of 0, whereas the synapse works exactly same as in the conventional model (i.e., Eq. (1)) when a certain element in C_W takes the value of 1. With this property, switching the values of C_W and C_K allows to model the synaptic pruning process intuitively (Fig. 1, top left).

The cost function L for this network was derived by integrating Eq. (6) with respect to neural activity x and time τ (Fig. 1, middle left and bottom left):

$$L=\int_0^t\left\{\begin{pmatrix}x(\tau)\\ \bar{x}(\tau)\end{pmatrix}^{T}\left\{\ln\begin{pmatrix}x(\tau)\\ \bar{x}(\tau)\end{pmatrix}-\left\{C_W^{\ddagger}\odot\ln\begin{pmatrix}\widehat{W}_1 & \widehat{\overline{W}_1}\\ \widehat{W}_0 & \widehat{\overline{W}_0}\end{pmatrix}\right\}\begin{pmatrix}o(\tau)\\ \bar{o}(\tau)\end{pmatrix}-\left\{C_K^{\ddagger}\odot\ln\begin{pmatrix}\widehat{K}_1 & \widehat{\overline{K}_1}\\ \widehat{K}_0 & \widehat{\overline{K}_0}\end{pmatrix}\right\}\begin{pmatrix}x(\tau-\Delta\tau)\\ \bar{x}(\tau-\Delta\tau)\end{pmatrix}-\begin{pmatrix}\phi_1\\ \phi_0\end{pmatrix}\right\}+\alpha\right\}d\tau+\mathscr{O}(1), \tag{7}$$

where the double dagger $\ddagger$ on C denotes the block matrix of C, $C^{\ddagger}:=\begin{pmatrix}C & C\\ C & C\end{pmatrix}$; α denotes the integration constant that is independent of x, W, and K but can depend on C_W and C_K; $\phi_l := h_l-(C_W\odot\ln\widehat{\overline{W_l}})\overrightarrow{1}-(C_K\odot\ln\widehat{\overline{K_l}})\overrightarrow{1}$ $(l=1,0)$. The details of the derivations of Eq. (7) (i.e. integration of Eq. (6)) are provided in Appendix A.

The differences between Eq. (2) and (7) are these three variables: C_W, C_K and α. Therefore, adding variables corresponding to these three to Eq. (5) yields variational free energy equivalent to Eq. (7) (Fig. 1, bottom left and bottom right):

$$F=\sum_{\tau=1}^{t}\left\{\mathbf{s}_\tau\cdot(\ln\mathbf{s}_\tau-\mathbf{c}_A^{\ddagger}\odot\ln\mathbf{A}\cdot o_\tau-\mathbf{c}_B^{\ddagger}\odot\ln\mathbf{B}^{\dagger}\cdot\mathbf{s}_{\tau-1}-\ln D)+Z\right\}+\mathscr{O}(\ln t), \tag{8}$$

where $\mathbf{c}_A$, $\mathbf{c}_B$ and Z are the posterior expectations and a normalization factor, which correspond to C_W, C_K and α, respectively. We describe the details of the free energy and confirm its formal correspondence to L in the next subsection.

Subsequently, a POMDP generative model that yields variational free energy represented in Eq. (8) was reverse engineered, as follows (Fig. 1, bottom right and middle right):

$$P(o_{1:t},\ s_{1:t},\theta)=P(A)P(B)P(C_A)P(C_B)\prod_{\tau=1}^{t}P(o_\tau|s_\tau,A,\ C_A)P(s_\tau|s_{\tau-1},B,C_B). \tag{9}$$

Here and throughout, $\theta:=\{A,B,C_A,C_B\}$ denotes a set of model parameters. Equation (9) presents a POMDP model that generates N_o binary observations $o_\tau := \left(o_\tau^{(1)},\ldots,o_\tau^{(N_o)}\right)^{\mathrm{T}}$ from N_s binary hidden states $s_\tau := \left(s_\tau^{(1)},\ldots,s_\tau^{(N_s)}\right)^{\mathrm{T}}$ through the likelihood mapping A and its connectivity C_A in the form of a categorical distribution, $P(o_\tau|s_\tau,\ A,\ C_A)=$

$\mathrm{Cat}\left(A^{\odot C_A^{\ddagger}} s_\tau\right)$. The superscript $\odot$ on A indicates the element-wise power. The hidden states s_τ depend on the previous states $s_{\tau-1}$, following the state transition matrix B and its connectivity C_B, $P(s_\tau|s_{\tau-1}, B, C_B) = \mathrm{Cat}\left(B^{\odot C_B^{\ddagger}} s_{\tau-1}\right)$. The new parameters $C_A \in \{1,0\}^{N_o \times N_s}$ and $C_B \in \{1,0\}^{N_s \times N_s}$ express the connectivity structure of graphical models. Owing to the factorial property of A, $C_{Aij} = 0$ means the reduction (i.e., elimination or pruning) of $A^{(i,j)}$ from the generative model, and the same applies to B (see Appendix B for further details). If $C_{Aij} = 1$ holds for all connectivities, Eq. (9) is reduced to a conventional full model without model reduction (Eq. (3)). The prior distributions of A, B and C_A, C_B follow the Dirichlet distributions $P(A) = \mathrm{Dir}(a)$, $P(B) = \mathrm{Dir}(b)$ and categorical distributions $P(C_A) = \mathrm{Cat}(c_A)$, $P(C_B) = \mathrm{Cat}(c_B)$, respectively. The details of the notations are provided in Appendix B. In the next subsection, we confirm that (Eq. (9)) is a biologically plausible generative model under which the neural network in Eq. (6) operates.

2.3. Variational free energy and belief update rule

Variational Bayesian inference is the process of optimizing an approximate posterior belief $Q(s_{1:t}, \theta)$ based on a sequence of observations by minimizing the variational free energy as a tractable proxy for surprise minimization. Based on the mean-field approximation, the posterior distribution is expressed as follows:

$$Q(s_{1:t}, \theta) = Q(A)Q(B)Q(C_A)Q(C_B)\prod_{\tau=1}^{t} Q(s_\tau). \tag{10}$$

The marginal posteriors about A, B, C_A, C_B, s_τ follow the Dirichlet distributions $Q(A) = \mathrm{Dir}(\mathbf{a})$, $Q(B) = \mathrm{Dir}(\mathbf{b})$ and categorical distributions $Q(C_A) = \mathrm{Cat}(\mathbf{c}_A)$, $Q(C_B) = \mathrm{Cat}(\mathbf{c}_B)$, $Q(s_\tau) = \mathrm{Cat}(\mathbf{s}_\tau)$, respectively. Here and throughout, the bold case variables (i.e., $\mathbf{c}_A$, $\mathbf{c}_B$, $\mathbf{s}_\tau$, $\mathbf{A}$, $\mathbf{B}$) indicate the posterior expectations, while $\mathbf{a}$ and $\mathbf{b}$ denote the concentration parameters (Table 1). The analytical expression for the variational free energy under the abovementioned generative model and posterior beliefs is given as follows:

$$F(o_{1:t},\ Q(s_{1:t}, \theta)) = \sum_{\tau=1}^{t} \left\{\mathbf{s}_\tau \cdot \left(\ln \mathbf{s}_\tau - \mathbf{c}_A^{\ddagger} \odot \ln \mathbf{A} \cdot o_\tau - \mathbf{c}_B^{\ddagger} \odot \ln \mathbf{B}\mathbf{s}_{\tau-1}\right) + \mathrm{E}_Q[\ln Z_{C_A}] + \mathrm{E}_Q[\ln Z_{C_B}]\right\} + \mathscr{D}_{\mathrm{KL}}[Q(\theta) \| P(\theta)], \tag{11}$$

where $\mathrm{E}_Q[\ln Z_{C_A}]$ and $\mathrm{E}_Q[\ln Z_{C_B}]$ denote the posterior expectations of normalization factors associated with $\mathbf{c}_A$ and $\mathbf{c}_B$, respectively; $\mathscr{D}_{\mathrm{KL}}[Q(\theta) \| P(\theta)]$ denotes the complexity scored by the Kullback–Leibler divergence between $Q(\theta)$ and $P(\theta)$. The details of the normalization factors are provided in Appendix B, and explicit forms of the posterior distribution and the free energy are provided in Appendix C. Applying a first-order approximation to A and B, $\mathrm{E}_Q[\ln Z]$ can be calculated as follows:

$$\mathrm{E}_Q[\ln Z_{C_A}] \approx \ln 2 \cdot \sum_{i=1}^{N_o}\left(1 - \sum_{j=1}^{N_s} \mathbf{c}_{A_{ij}}\right),\ \mathrm{E}_Q[\ln Z_{C_B}] \approx \ln 2 \cdot \sum_{i=1}^{N_s}\left(1 - \sum_{j=1}^{N_s} \mathbf{c}_{B_{ij}}\right). \tag{12}$$

The detailed derivations of this approximation are provided in Appendix D. The term $\mathrm{E}_Q[\ln Z_{C_A}] + \mathrm{E}_Q[\ln Z_{C_B}]$ is the explicit form of Z in Eq. (8). For Z in Eq. (8) to correspond to α in Eq. (7), it is necessary that Z is independent of $\mathbf{s}$, $\mathbf{A}$, and $\mathbf{B}$ because α is independent of x, W, and K. Eq. (12) confirms that this condition is met, meaning that BSyMP is plausible with the neural networks equipped with synaptic pruning.

Variational free energy minimization yields posterior beliefs about the hidden states and parameters. By solving the fixed point of the implicit gradient descent $\partial F/\partial \mathbf{s}_\tau = 0$ and $\partial F/\partial \boldsymbol{\theta} = O$ ($\boldsymbol{\theta} := \{\mathbf{a}, \mathbf{b}, \mathbf{c}_A, \mathbf{c}_B\}$), the update rules are derived as follows:

$$\mathbf{s}_\tau^{(n)} = \sigma\left(\sum_{i=1}^{N_o} \mathbf{c}_{Ain} \ln \mathbf{A}^{(i,n)} \cdot o_\tau^{(i)} + \sum_{j=1}^{N_s} \mathbf{c}_{Bnj} \ln \mathbf{B}^{(n,j)} \mathbf{s}_{\tau-1}^{(j)}\right), \tag{13}$$

$$\mathbf{a}^{(i,j)} = a^{(i,j)} + \mathbf{c}_{Aij} \sum_{\tau=1}^{t} o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}, \tag{14}$$

$$\mathbf{b}^{(i,j)} = b^{(i,j)} + \mathbf{c}_{Bij} \sum_{\tau=2}^{t} \mathbf{s}_\tau^{(i)} \otimes \mathbf{s}_{\tau-1}^{(j)}, \tag{15}$$

$$\mathbf{c}_{Aij} = \mathrm{sig}\left(\mathrm{sig}^{-1}\left(c_{Aij}\right) + \sum_{\tau=1}^{t} \left(\ln \mathbf{A}^{(i,j)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right) + \ln 2\right)\right), \tag{16}$$

$$\mathbf{c}_{Bij} = \mathrm{sig}\left(\mathrm{sig}^{-1}\left(c_{Bij}\right) + \sum_{\tau=2}^{t} \left(\ln \mathbf{B}^{(i,j)} \cdot \left(\mathbf{s}_\tau^{(i)} \otimes \mathbf{s}_{\tau-1}^{(j)}\right) + \ln 2\right)\right), \tag{17}$$

where $\otimes$ denotes the outer product operator, for example $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)} = o_\tau^{(i)}\left(\mathbf{s}_\tau^{(j)}\right)^{\mathrm{T}}$. The detailed derivations of the update rules are provided in Appendix E. Equations (16) and (17) are equivalent to the equations for the dynamics of C_W and C_K in the neural network formation that are derived from the fixed point of the gradient descent on L, $\partial L/\partial C_W = O$ and $\partial L/\partial C_K = O$. That is, Eqs. (16) and (17) are plausible synaptic pruning rules for the canonical neural network.

Table 1
Glossary of variables.

	Variables in neural networks	Variables in Bayesian inference	
Sensory inputs	$o(\tau) := (o_1(\tau), \ldots, o_{N_o}(\tau))^{\mathrm{T}} \in \{0,1\}^{2 \times N_o}$	$o_\tau := \left(o_\tau^{(1)}, \ldots, o_\tau^{(N_o)}\right)^{\mathrm{T}} \in \{0,1\}^{2 \times N_o}$	Observations
Neural activity	$x(\tau) := (x_1(\tau), \ldots, x_{N_x}(\tau))^{\mathrm{T}} \in [0,1]^{N_x}$	$\mathbf{s}_\tau := \left(\mathbf{s}_\tau^{(1)}, \ldots, \mathbf{s}_\tau^{(N_s)}\right)^{\mathrm{T}} \in [0,1]^{2 \times N_s}$	State posterior
Feedforward synaptic strengths	$W \in \mathbb{R}^{N_x \times N_o}$	$\mathbf{a} \in \left(\mathbb{R}^{2\times 2}\right)^{N_o \times N_s}$	Likelihood mapping posterior
Recurrent synaptic strengths	$K \in \mathbb{R}^{N_x \times N_x}$	$\mathbf{b} \in \left(\mathbb{R}^{2\times 2}\right)^{N_s \times N_s}$	State transition posterior
Survival probability of feedforward synapses	$C_W \in [0,1]^{N_x \times N_o}$	$\mathbf{c}_A \in [0,1]^{N_o \times N_s}$	Connectivity posterior for A
Survival probability of recurrent synapses	$C_K \in [0,1]^{N_x \times N_x}$	$\mathbf{c}_B \in [0,1]^{N_s \times N_s}$	Connectivity posterior for B
Adaptive firing thresholds	$h \in \mathbb{R}^{N_x}$	$D \in [0,1]^{2 \times N_s}$	State prior (constant)

Note: The variables in bold for the Dirichlet distributions (i.e., $\mathbf{a}$ and $\mathbf{b}$) indicate the concentration parameters, and those for categorical distributions (i.e., $\mathbf{c}_A$, $\mathbf{c}_B$, $\mathbf{s}_\tau$) indicate the posterior expectations of the corresponding variables in italics.

In short, the connectivity parameters C_A and C_B provide an implementation of the Bayesian model selection that enables switching between the presence (ON) and absence (OFF) of each element of matrices A and B. As shown in Fig. 1, this model reduction is formally equivalent to synaptic pruning in neural network formation, which is mediated by C_W and C_K in Eq. (6). Table 1 summarizes the correspondence between variables in neural networks and those in variational Bayes.

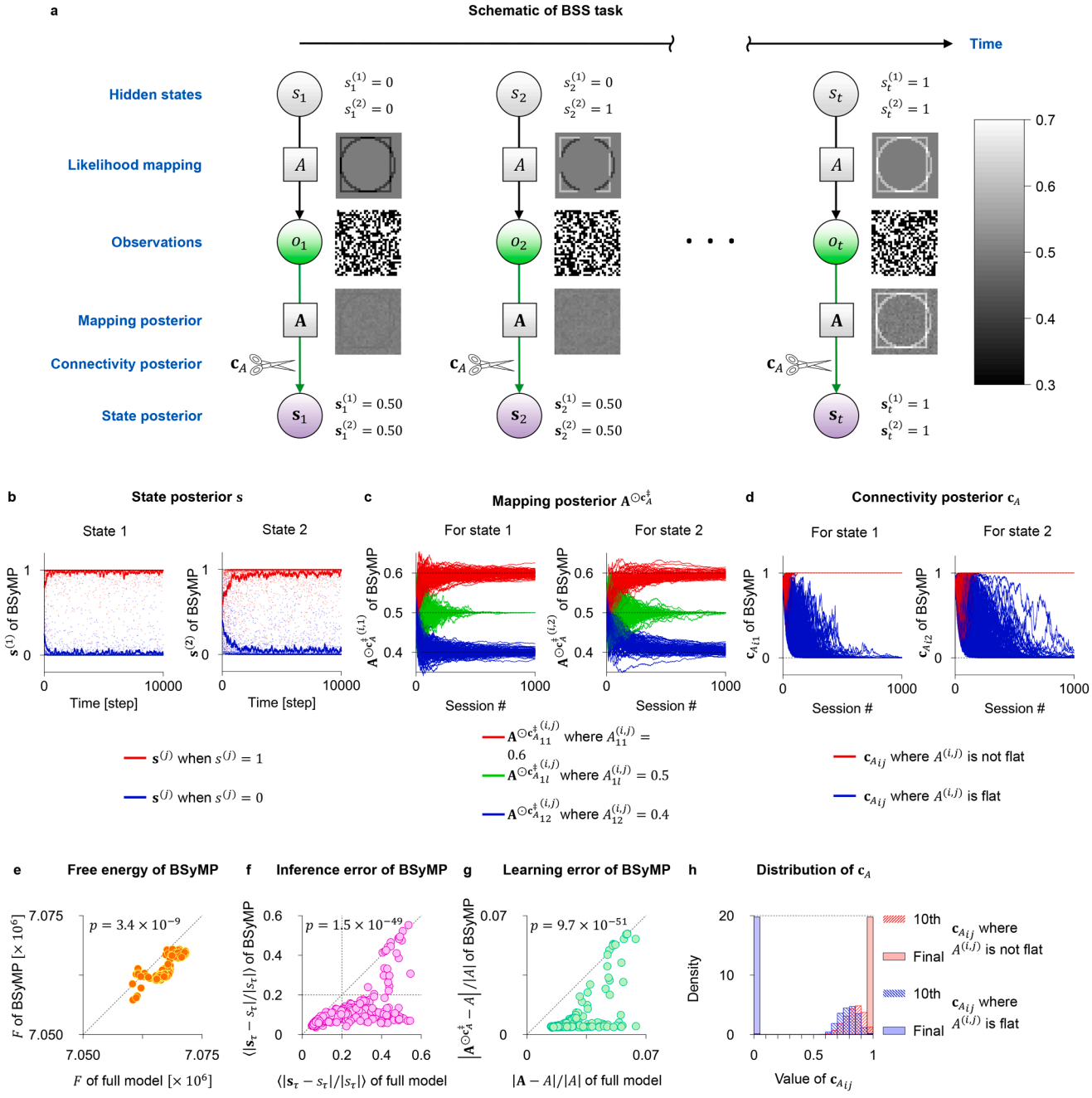


Fig. 2. Performance of BSyMP agent on BSS task. (**a**) A schematic of the task, illustrating a generative process with one likelihood mapping out of three. Two hidden sources s_τ randomly take the value of 1 (ON) or 0 (OFF). Binary observations o_τ are stochastically generated from s_τ through the likelihood mapping A, $P(o_\tau|s_\tau, A) = \mathrm{Cat}(As_\tau)$. The three vertically aligned images depict As_τ, o_τ, and $\mathbf{A}s_\tau$ in order from top to bottom. (**b**) The posterior beliefs about two states during a single simulation. The red and blue dots indicate $\mathbf{s}_\tau^{(j)}$ when the true hidden state $s_\tau^{(j)}$ is 1 or 0, respectively. The red and blue trajectories indicate the moving averages of the red and blue dots with a window length of 100 time points. (**c**) The posterior beliefs about the likelihood mapping during a single simulation. The elements in the posterior beliefs are grouped and colored according to the value of the corresponding elements in the true mapping, that is, 0.6 (red), 0.4 (blue), or 0.5 (green). (**d**) The posterior beliefs about the connectivity during a single simulation. The elements in the posterior beliefs are grouped and colored according to whether the corresponding submatrix in the true mapping is flat (blue; to be reduced) or not (red; to be retained). (**e**) A comparison of the free energy between BSyMP and the full model. In (**e**), (**f**), and (**g**), each point represents data obtained in a single simulation. Throughout this work, the p value was obtained by the two-sided Wilcoxon signed-rank test for the pair. (**f**) A comparison of the state inference error between BSyMP and the full model. The error was evaluated by the mean of the Euclidean norm ratio $|\mathbf{s}_\tau - s_\tau|/|s_\tau|$ in the test session. (**g**) A comparison of the parameter learning error between BSyMP and the full model. The error was evaluated by the Frobenius norm ratio $|\mathbf{A}^{\odot \mathbf{c}_A^\ddagger} - A|/|A|$ for BSyMP and $|\mathbf{A} - A|/|A|$ for the full model at the end of the training sessions. (**h**) The distribution of the posterior beliefs about the connectivity. All the data at the 10th and final training sessions of 300 simulations were counted in 0.05 intervals. Please refer to Appendix G for further details.

2.4. Properties of connectivity parameters

Equation (16) can be viewed as the dynamics of synaptic pruning in neural network formation. The probability of pruning is controlled by the similarities between the current synaptic strength ($\ln\mathbf{A}^{(i,j)}$) and its change, as calculated by the Hebbian rule ($o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$) (Hebb, 1949). Because the values in $\ln\mathbf{A}^{(i,j)}$ are all negative and the values in $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$ are all positive, their product $\ln\mathbf{A}^{(i,j)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right)$ is a negative value. The sum of their product over time $\sum_{\tau=1}^{t} \left(\ln\mathbf{A}^{(i,j)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right)\right)$ provides the empirical estimate of the negative cross-entropy between $\mathbf{A}^{(i,j)}$ and $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$. Therefore, the summation in Eq. (16) becomes large negative when $\mathbf{A}^{(i,j)}$ and $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$ are considerably different, leading to the pruning of the corresponding connection, and becomes positive when these are similar, leading to the maintenance of the connection.

In addition, mathematical analyses show that an element of the connectivity parameter $\mathbf{c}_{A^{ij}}$ decreases and converges to 0 when the corresponding element of observation $o_\tau^{(i)}$ is randomly determined and independent of all elements of s_τ if the likelihood mapping posterior $\mathbf{A}^{(i,j)}$ and the mean state posterior $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]$ are sufficiently flat (refer to Appendix F for further details). This means that BSyMP is guaranteed to prune unnecessary connections in the generative model as long as the prior of the likelihood mapping and the mean state posterior are unbiased. In terms of neural networks, the assumption of $\mathbf{A}^{(i,j)}$ being sufficiently flat is associated with the absence of giant synapses, which would be satisfied in the early developmental stage. Moreover, the assumption of $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]$ being sufficiently flat can be satisfied by homeostasis (homeostatic plasticity) mechanisms in the brain, which maintain the average excitatory level of neurons within an appropriate range.

3. Results

3.1. Numerical simulation 1: blind source separation

The performance of BSyMP was evaluated using a blind source separation (BSS) task, which is a standard benchmark to investigate causal inference in neural networks (Brown et al., 2001; Choi et al., 2004; Isomura et al., 2015, 2023). The task was designed so that the performance was expected to improve with appropriate model reduction. This BSS task was analogous to a visual shape recognition task in which sources 1 and 2 displayed a low-contrast circle and rectangle, respectively, on a 32 × 32 pixel screen with a certain fidelity following a likelihood mapping (Fig. 2a). Each pixel on the display took either 0 or 1 stochastically, where some parts of the shapes overlapped. Thus, the agent needed to disentangle the input signals to perceive the hidden states. The field of view of the agent comprised 1024-dimensional observations, and the states of the two sources (ON/OFF) changed randomly. This setup was effectively expressed as a generative model without matrix B; thus, the agents optimized only $\mathbf{s}_\tau$, $\mathbf{a}$, and $\mathbf{c}_A$.

Three agent types (full model, BMR, and BSyMP) were trained with identical input sequences. The agents optimized $\mathbf{s}_\tau$ at every time point; conversely, they optimized $\mathbf{a}$ and $\mathbf{c}_A$ at every 10 time points, which we refer to as a training session. One simulation comprised 1000 training sessions and a subsequent test session. In every test session, the agents inferred $\mathbf{s}_\tau$ for 1000 time-point-length test data that were newly generated from the same likelihood mapping used in the training sessions. We prepared three likelihood mappings, generating inputs that differed in the size of the circle and rectangle and the overlap between them. The simulation was repeated 100 times for each mapping (i.e., 300 simulations in total). Further details are provided in Appendix G and the source code.

The agents with BSyMP successfully self-organized to infer the hidden sources as a result of successful learning of posterior beliefs about the likelihood mapping (Fig. 2b and 2c). The result was underwritten by an appropriate model reduction, as depicted by the efficient reduction of the redundant connections (Fig. 2d). When the source and observation were independent of one another, BSyMP reliably reduced the connectivity between them to 0 (Fig. 2d; blue lines). This result was consistent with the mathematical analysis described in Section 2.4 and Appendix D, Appendix F. Fig. 2d further shows that the connectivity was retained if the source and the observation were not independent (red lines), indicating that no over-pruning occurred in this task. Owing to these desirable properties of the connectivity parameters, the variational free energy of the BSyMP agent was smaller than that of the full model agent (Fig. 2e), and the inference and learning performance of the BSyMP agent in this BSS task significantly exceeded those of the full model agent (Fig. 2f-h). The full model refers to the conventional model which does not involve the connectivity parameter C, whose generative model is given by Eq. (3). For ease of discussion, we labeled a simulation as successful if the state inference error in the test session was <20 %, and as a failure otherwise (Fig. 2f; vertical and horizontal dashed lines). Under this criterion, the success rate of the full model agent was 45.0 % (135/300), whereas that of the BSyMP agent was 92.3 % (277/300). These results demonstrate that BSyMP achieved appropriate model reduction and significant effects on the performance of the BSS task.

Almost no synaptic genesis occurred in the simulations despite that no constraint was involved to prevent synaptic genesis in our model. The mean number of synaptic genesis was only 1.85 ± 1.90 out of 2048 synapses (mean ± standard deviation in $n = 300$ simulations), where the occurrence of synaptic genesis was counted when a connectivity ($\mathbf{c}_{A^{ij}}$) that had been <0.05 at any time point in the simulations subsequently became larger than 0.5. This was arguably because in the simulations the prior (initial) connection probability was set to high (0.9 ± 0.05) in accordance with the characteristics of the brain during the development stage, and BSyMP reduced $\mathbf{c}_{A^{ij}}$ for unnecessary connections efficiently and kept them approximately zero as described in Methods Section 2.4 and Appendix F. Although $\mathbf{c}_{A^{ij}}$ for necessary connections decreased at the beginning of training owing to the stochasticity, it rarely became <0.05. These properties made synaptic genesis unlikely to occur in our setting.

Subsequently, BSyMP was compared with a standard Bayesian model reduction method, referred to as BMR. BMR compares the free energy of the learned full model with that of the reduced version of the model at a small computational cost. Thus, BMR is a post hoc model reduction method that is based on an already acquired model.

Although the BMR agent decreased the variational free energy more than the BSyMP agent, the performance of the BMR agent was close to that of the full model and did not exceed that of the BSyMP agent (Fig. 3a-c). The success rate of BMR was 46.7 % (140/300), which was only 1.7 % higher than that of the full model, whereas the rate of BSyMP was twice as high as that of the full model. The inconsistency between the free energy and the performance was arguably because BMR decreased the complexity term $\mathscr{D}_{\mathrm{KL}}[Q(A)||P(A)]$ in the free energy by heuristically altering priors (Friston et al., 2017), but this free energy reduction had little contribution to the performance in this setting. By contrast, BSyMP could prune synapses by evaluating the likelihood through its construction. Consistent with the performance, the accuracy of model reduction by BMR is worse than that of BSyMP (Fig. 3d).

We attribute the underperformance of BMR to the post hoc manner of its model reduction. This hypothesis was supported by the comparison of parameter trajectories. Fig. 3e illustrates the typical outcomes of three approaches in a simulation. Owing to stochasticity in the generative processes and priors, the agent with the full model often (in 55 % of the simulations) failed to learn the likelihood mapping (Fig. 3e, left panel). BMR could reduce the redundant elements but could not change the learning process because it operates in a post hoc manner (Fig. 3e, center panel). By contrast, as BSyMP adopts online reduction during learning, it could learn the likelihood mapping based on the state inference that was improved by the model reduction, resulting in its reliable success

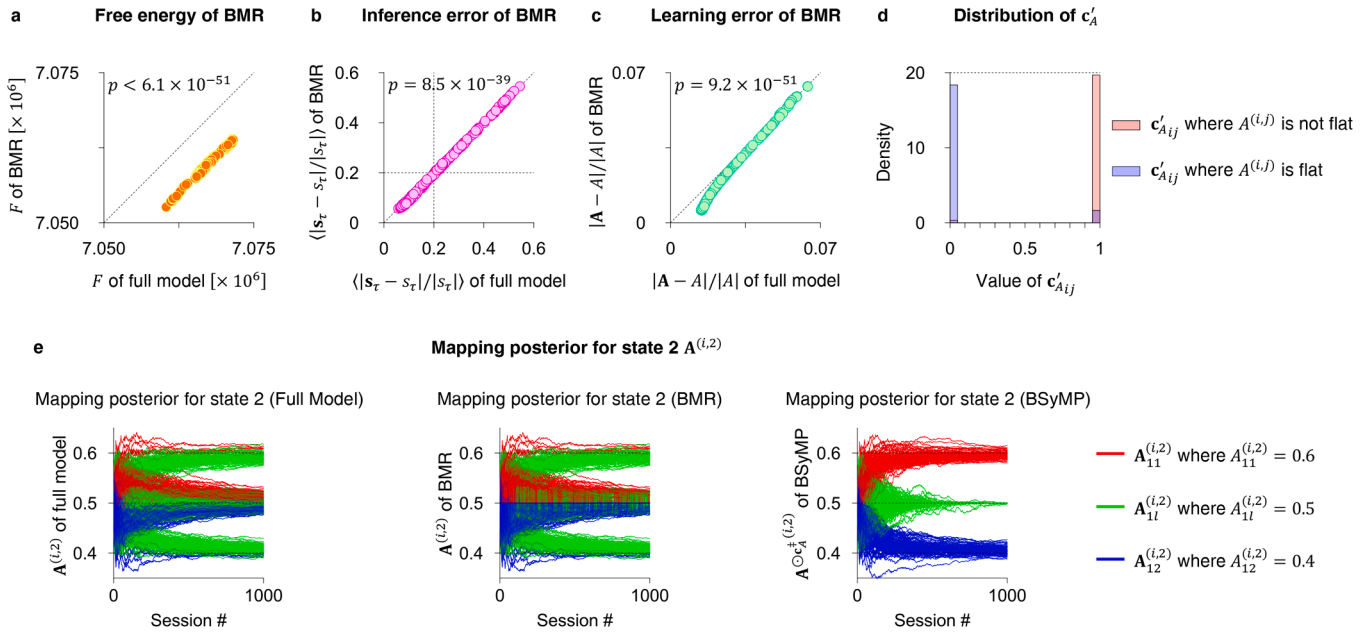


Fig. 3. Comparison of BSyMP with BMR on BSS task. (**a**) A comparison of the free energy between BMR and the full model. (**b**) A comparison of the state inference error between BMR and the full model. (**c**) A comparison of the parameter learning error between BMR and the full model. (**d**) The distribution of the BMR's counterpart, $\mathbf{c}'_A$, to the connectivity parameter in BSyMP, $\mathbf{c}_A$, at the end of the training sessions. Here, $\mathbf{c}'_A$ has the same number of elements as $\mathbf{c}_A$, and each element in $\mathbf{c}'_A$ takes a binary value, indicating whether the corresponding submatrix of the mapping posterior is reduced (0) or not (1). (**e**) The posterior beliefs about the likelihood mapping regarding state 2 during a single simulation obtained by three methods. Note that the time course of the posterior beliefs of a BMR agent was calculated by performing BMR at every session.

(Fig. 3e, right panel). While naive BMR operates in the post-hoc manner, BMR can be applied to operate online (Friston et al., 2017). However, the online version of BMR performed the same level as the post-hoc BMR, because of its heuristics (Appendix H). We also analyzed BMR and compared it with BSyMP mathematically. The analysis shows that BMR reduces the sub-matrix $A^{(ij)}$ when it is sufficiently flat, whereas BSyMP reduces $A^{(ij)}$ based on its similarity to $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$ (Appendix I). In addition, the mathematical comparison shows that BSyMP provides Bayes-optimal model reduction, whereas BMR provides heuristic model reduction (Appendix J). In summary, we demonstrated that BSyMP successfully eliminated connections independent of the sources and improved the performance of the BSS task more significantly than related methods.

3.2. Numerical simulation 2: rule learning

The performance of BSyMP was further evaluated using a rule-learning task in which the agents learned the transition rule behind the visual inputs in addition to BSS. The agents received observations comprising 32 × 96 pixels, which was analogous to three screens. Four sources on each screen stochastically displayed a circle, rectangle, x, and plus, respectively (Fig. 4a). In other words, the agents received 3072 binary observations that depend on the states of the 12 sources (ON/OFF). The states changed according to the transition rule described by matrix B. The agents had to learn the rule to predict the next state of the sources.

We trained the three agent types in the same manner as in the BSS task. The agents optimized $\mathbf{s}_\tau$ at every time point and optimized $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}_A$, and $\mathbf{c}_B$ at every 100 time points during the training sessions. In this case, we provided the agents with accurate and strong prior beliefs about the likelihood mapping A to focus on the learning in matrix B, thereby ensuring accurate state inference after obtaining the observations. One simulation comprised 100 training sessions and a subsequent 1000 time-point-length test session in which the prediction error for the upcoming states was evaluated. We prepared three transition rules, and the simulation was repeated 100 times for each rule (i.e., 300 simulations in total). Further details are provided in Appendix K and the source code.

The BSyMP agents scored a smaller variational free energy than the full model agent and successfully predicted the upcoming hidden states by learning the posterior beliefs about the transition rules (Fig. 4b-d). The success was underwritten by the appropriate pruning of redundant recurrent connections between the state posteriors (Fig. 4e). The prediction and learning performance of the BSyMP agent in the rule-learning task significantly exceeded those of the full model agent (Fig. 4c and d) owing to the desirable properties of the connectivity parameters. For ease of discussion, we labeled a simulation as successful if the state prediction error in the test session was <10 %, and as a failure otherwise (Fig. 4c). Under this criterion, the success rate of the full model agent was 9.0 % (27/300), whereas that of the BSyMP agent was 82.3 % (247/300). These results show that BSyMP achieved appropriate model reduction and significant effects on the performance of the rule-learning task.

Subsequently, BSyMP was compared with BMR. As in the BSS task, although BMR decreased the variational free energy more than BSyMP owing to the decrease in complexity (Fig. 4f), and the performance of the BMR agent exceeded that of the full model agent (Fig. 4g and h), the success rate of the BMR agent, 58.7 % (176/300), was less than that of BSyMP. This performance difference could not be attributed to the difference in the manner of model reduction (i.e., online or post hoc) because the model reduction had no significant effect on learning matrix B in this rule-learning task, owing to accurate state inference, unlike in the BSS task. Nevertheless, pruning unnecessary connections is more challenging in this task than in the BSS task because some sources can be correlated with others despite the absence of direct connections. For example, in the rule shown in Fig. 4a, the state of the rectangle on screen 1 at time τ is negatively correlated with the state of the plus on screen 1 at time $\tau + 1$. The blue bars in Fig. 4e and 4i show that BSyMP was better at pruning these "false positive connections" than BMR, which can

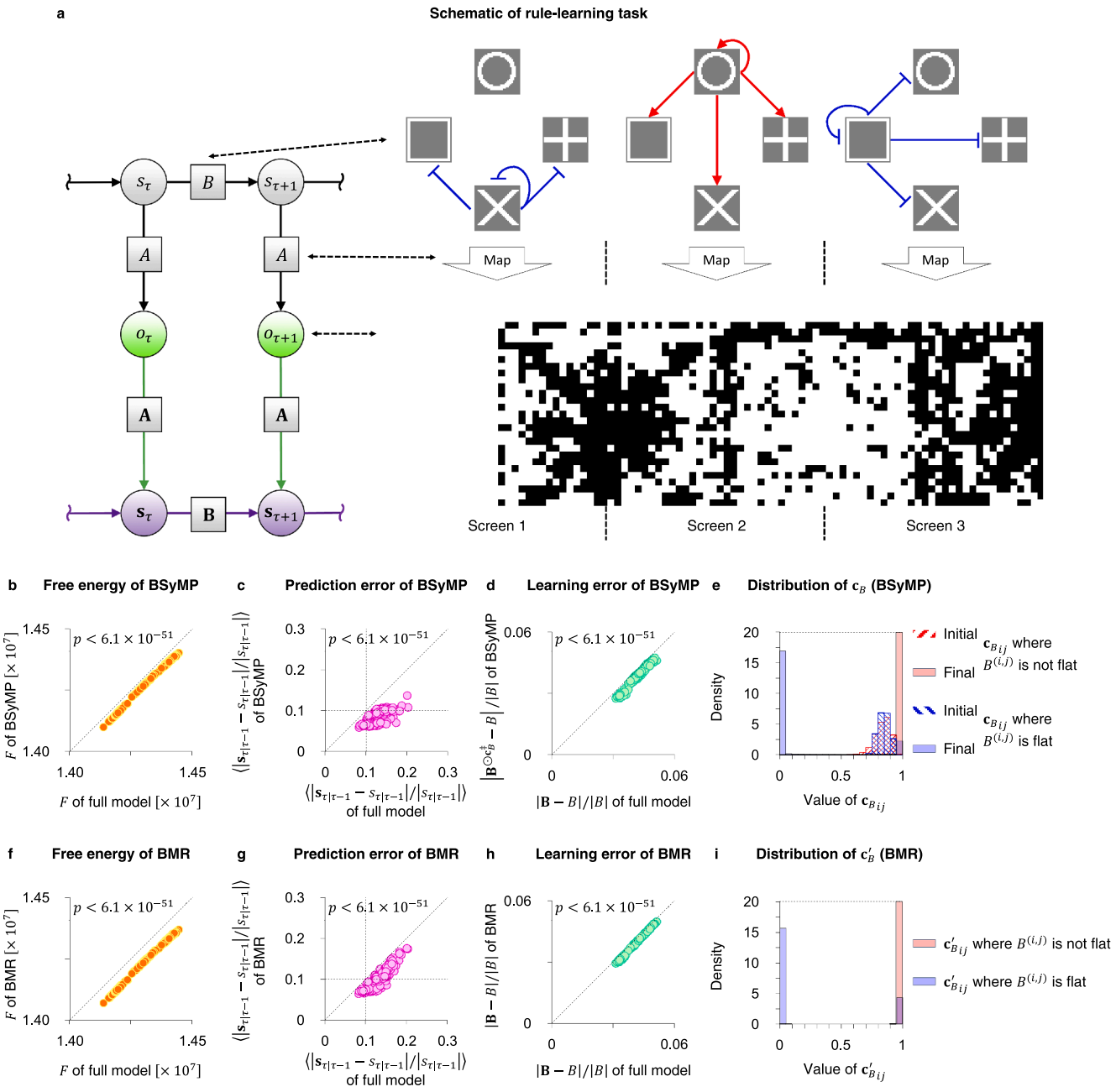


Fig. 4. Performance of BSyMP agent on rule-learning task. (**a**) A schematic of the task. The factor graph on the left side illustrates a generative process. Images on the right side depict one of three transition rules and one example of observations at a moment. The states of the sources stochastically change depending on the states at the previous time point. The red arrows from one source $s^{(j)}$ to another source $s^{(i)}$ indicate the existence of the positive connection from $s^{(j)}_{\tau-1}$ to $s^{(i)}_{\tau}$, whereas the blue T-shaped lines indicate the negative connection. Each source displays a shape in one of three 32 × 32-pixel screens and has no effect on the other 32 × 64 pixels. The agent optimizes the posterior beliefs about s_τ, A, B, C_A, and C_B. (**b**) A comparison of the free energy between BSyMP and the full model. (**c**) A comparison of the state prediction error between BSyMP and the full model. The error was evaluated by the mean of the normalized Euclidean distance between the actual probability of the state and agent's prediction $|\mathbf{s}_{\tau|\tau-1} - s_{\tau|\tau-1}|/|s_{\tau|\tau-1}|$ in the test session. Please refer to Appendix K for the definitions of $\mathbf{s}_{\tau|\tau-1}$ and $s_{\tau|\tau-1}$. (**d**) A comparison of the rule-learning error between BSyMP and the full model. The error was evaluated by the Frobenius norm ratio $|\mathbf{B}^{\circ\mathbf{c}_B^{\ddagger}} - B|/|B|$ for BSyMP and $|\mathbf{B} - B|/|B|$ for the full model at the end of the training sessions. (**e**) The distribution of the posterior beliefs about the connectivity parameter. (**f**) A comparison of the free energy between BMR and the full model. (**g**) A comparison of the state prediction error between BMR and the full model. (**h**) A comparison of the rule-learning error between BMR and the full model. (**i**) The distribution of the BMR's counterpart, $\mathbf{c}'_B$, to the connectivity parameter in BSyMP, $\mathbf{c}_B$, at the end of the training sessions.

explain the outperformance of BSyMP in rule-learning tasks. In summary, we demonstrated that BSyMP successfully eliminated unnecessary connections between the sources and improved the rule-learning performance to a greater extent than the previously developed method.

4. Discussion

We derived the biologically plausible synaptic pruning rule, BSyMP (Eqs. (16) and (17)), from the cost function minimization based on the equivalence between the dynamics of the canonical neural networks and variational Bayesian inference. Equation (16) states that the probability of pruning depends on the similarities between the current synaptic strength ($\ln \mathbf{A}^{(i,j)}$) and its change calculated by the Hebbian rule ($o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$) during the latest session. The Hebbian plasticity in the brain can be achieved by long-term potentiation (LTP) and depression (LTD), and its molecular mechanisms are well studied (Malenka & Bear, 2004). The Hebbian term in Eq. (16) can be encoded in the molecules in the postsynaptic neuron (Nicoll, 2017; Scannevin & Huganir, 2000), while the values of synaptic strengths lie in the density of AMPA receptors in the synaptic plasma membrane. Thus, it is possible that the molecular mechanisms in the postsynaptic neuron compute the correlation between $\ln \mathbf{A}^{(i,j)}$ and $o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}$ to determine which synapses are pruned. Experimentally observed similarities between LTP, LTD, and synaptic pruning—in their physiological properties and molecular mechanisms—further support the biological plausibility of our model, highlighting its potential to guide the experimental targets to shed light on the mechanisms underlying synaptic connectivity control (Helias et al., 2008; Le Bé & Markram, 2006; Lüscher et al., 2000).

In terms of the biological plausibility of BSyMP, synaptic plasticity occurs depending on the neural activity, and synaptic weights change in the timescale of dozen seconds. In accordance with these empirical observations, our model updates synaptic weights based on Hebbian product of two successive variables accumulated during a trial, as we assume a trial is finished within dozen seconds. Thus, the likelihood and transition matrices and their corresponding connectivity can be learned through Hebbian plasticity that depends on two variables in time τ and $\tau - 1$ in a biologically plausible manner without relying on stock of long sequences.

One can contrast BSyMP with earlier pruning methods. Pruning connections with small weight is the most basic and widely used rule to prune neural networks (Hagiwara, 1994; Han et al., 2015, 2016; Hoefler et al., 2021; Kang et al., 2008). Our analysis shows that BMR constitutes a family of weight-base pruning rules (please refer to Appendix I for details). In addition to weight-based methods, pruning methods based on the activity of neurons exist (Scholl et al., 2021; Sietsma & Dow, 1991; Sun et al., 2015). BSyMP utilizes temporal consistency to organize the network structure by comparing synaptic weight (i.e., the history of neural activity in the past) with neural activity in the latest session. This allows the neural network to spontaneously detect changes in the statistical features of neural activity. This would be useful for reorganizing the network structure flexibly to adapt to environmental changes, enabling BSyMP to outstand in the activity-based pruning rules. This might be a mechanistic explanation of the dynamics of the network structure in the brain (Ziv & Brenner, 2018). Moreover, most pruning methods, including BMR, prune synapses in a binary manner, and their processes involve some heuristics. BSyMP, by contrast, computes the probability of the presence of synapses and prunes synapses in a continuous (i.e., differentiable) manner, thereby enabling stochastic gradient descent. This rule is formally derived from variational free energy minimization, achieving more accurate pruning than the binary methods. This is a great advantage over heuristic methods. Appendix J delves into this contrast analytically.

Biological organisms utilize structural learning to identify the structure of their environment (Braun et al., 2010; Lynn & Bassett, 2020; Tenenbaum et al., 2011). Structure learning facilitates generalization by transferring the structures obtained in the past to newly encountered environments (Collins & Frank, 2013). Our work suggests that synaptic pruning plays a vital role in structure learning in the brain. This would be useful for designing next-generation artificial intelligence with high data efficiency and robustness and for understanding the mechanisms underlying the generalizability of the brain and its attenuation caused by psychiatric disorders (Kasai et al., 2021).

5. Conclusion

In this work, we derived a reliable online Bayesian model selection scheme, BSyMP, that can be implemented by canonical neural networks equipped with synaptic pruning. Mathematical analyses revealed that BSyMP is guaranteed to prune unnecessary connections in the generative model. Numerical simulations of BSS and rule-learning tasks demonstrated the advantage of BSyMP over the conventional BMR scheme. These findings suggest that synaptic pruning can be a possible neuronal substrate underlying structure learning and generalizability in the brain.

CRediT authorship contribution statement

Ukyo T. Tazawa: Data curation, Software, Writing – original draft, Methodology, Formal analysis, Writing – review & editing, Visualization, Investigation, Conceptualization. **Takuya Isomura:** Writing – review & editing, Resources, Formal analysis, Methodology, Conceptualization, Supervision, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Detailed derivation of the neural network cost function from the neural activity dynamics

The neural network cost function (Eq. (7)) was derived so that the gradient descent on the function yields the neural activity dynamics (Eq. (6)). As we consider that the gradient descent on L should yield Eq. (6), i.e., $\dot{x}(\tau) \propto -\frac{\partial}{\partial x}\frac{\partial L}{\partial \tau}$, the following relationship should hold by construction:

$$-\frac{\partial}{\partial x}\frac{\partial L}{\partial \tau} = -\mathrm{sig}^{-1}(x(\tau)) + \{C_W \odot (W_1 - W_0)\}o(\tau) + \{C_K \odot (K_1 - K_0)\}x(\tau - \Delta t) + h_1 - h_0. \tag{A.1}$$

Note that the right-hand side of Eq. (A.1) is the same as that of Eq. (6). Fixing all the elements of C_W and C_K to 1 in the following description confirms that Eq. (2) yields Eq. (1). Differentiating Eq. (7) with respect to x and τ provides the following:

$$\frac{\partial}{\partial x}\frac{\partial L}{\partial \tau} = \frac{\partial}{\partial x}\left\{\begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix}^T \left\{\ln\begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix} - \left\{C_W^{\ddagger} \odot \ln\begin{pmatrix} \widehat{W}_1 & \overline{\widehat{W}_1} \\ \widehat{W}_0 & \overline{\widehat{W}_0} \end{pmatrix}\right\}\begin{pmatrix} o(\tau) \\ \bar{o}(\tau) \end{pmatrix} - \left\{C_K^{\ddagger} \odot \ln\begin{pmatrix} \widehat{K}_1 & \overline{\widehat{K}_1} \\ \widehat{K}_0 & \overline{\widehat{K}_0} \end{pmatrix}\right\}\begin{pmatrix} x(\tau - \Delta\tau) \\ \bar{x}(\tau - \Delta\tau) \end{pmatrix} - \begin{pmatrix} \phi_1 \\ \phi_0 \end{pmatrix}\right\} + \alpha\right\}$$

$$= \begin{pmatrix} \overrightarrow{1} - \overrightarrow{1} \end{pmatrix}^T \left\{\ln\begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix} - \left\{C_W^{\ddagger} \odot \ln\begin{pmatrix} \widehat{W}_1 & \overline{\widehat{W}_1} \\ \widehat{W}_0 & \overline{\widehat{W}_0} \end{pmatrix}\right\}\begin{pmatrix} o(\tau) \\ \bar{o}(\tau) \end{pmatrix} - \left\{C_K^{\ddagger} \odot \ln\begin{pmatrix} \widehat{K}_1 & \overline{\widehat{K}_1} \\ \widehat{K}_0 & \overline{\widehat{K}_0} \end{pmatrix}\right\}\begin{pmatrix} x(\tau - \Delta\tau) \\ \bar{x}(\tau - \Delta\tau) \end{pmatrix} - \begin{pmatrix} \phi_1 \\ \phi_0 \end{pmatrix}\right\} + \begin{pmatrix} x(\tau) \\ \bar{x}(\tau) \end{pmatrix}^T \begin{pmatrix} 1/x(\tau) \\ -1/\bar{x}(\tau) \end{pmatrix} \tag{A.2}$$

For a term in the right-hand side of Eq. (A.2), the following equation holds by definition:

$$\begin{pmatrix} \overrightarrow{1} - \overrightarrow{1} \end{pmatrix}^T \left\{C_W^{\ddagger} \odot \ln\begin{pmatrix} \widehat{W}_1 & \overline{\widehat{W}_1} \\ \widehat{W}_0 & \overline{\widehat{W}_0} \end{pmatrix}\right\}\begin{pmatrix} o(\tau) \\ \bar{o}(\tau) \end{pmatrix} = C_W \odot (W_1 - W_0)o(\tau) + C_W \odot (\ln\overline{\widehat{W}_1} - \ln\overline{\widehat{W}_0})\overrightarrow{1}. \tag{A.3}$$

The similar equation holds for the term including $C_K^{\ddagger}$. Therefore, Eq. (A.2) can be transformed further as follows:

$$\frac{\partial}{\partial x}\frac{\partial L}{\partial \tau} = \ln x(\tau) - \ln\bar{x}(\tau) - C_W \odot (W_1 - W_0)o(\tau) - C_K \odot (K_1 - K_0)x(\tau - \Delta\tau) - (h_1 - h_0). \tag{A.4}$$

As $\ln x(\tau) - \ln\bar{x}(\tau) = \mathrm{sig}^{-1}(x(\tau))$ holds, Eq. (A.1) is the negative of Eq. (A.4). Therefore, the gradient descent on Eq. (7) is confirmed to yield Eq. (6).

Appendix B. Detailed description of generative model

The explicit form of this generative model is expressed as follows:

$$P(o_{1:t}, s_{1:t},\ \theta) = P(A)P(B)P(C_A)P(C_B)\prod_{\tau=1}^{t} P(o_\tau|s_\tau, A,\ C_A)P(s_\tau|s_{\tau-1}, B, C_B)$$

$$= \prod_{i=1}^{N_o}\prod_{j=1}^{N_s} P\left(A^{(ij)}\right)P\left(C_{Aij}\right)\cdot \prod_{i=1}^{N_s}\prod_{j=1}^{N_s} P\left(B^{(ij)}\right)P\left(C_{Bij}\right)\cdot$$

$$\prod_{\tau=1}^{t}\left\{\prod_{i=1}^{N_o}\frac{\prod_{j=1}^{N_s} P\left(o_\tau^{(i)}\middle|s_\tau^{(j)}, A^{(ij)}, C_{Aij}\right)}{Z_{C_A}^{(i)}}\cdot \prod_{i=1}^{N_s}\frac{\prod_{j=1}^{N_s} P\left(s_\tau^{(i)}\middle|s_{\tau-1}^{(j)}, B^{(ij)}, C_{Bij}\right)}{Z_{C_B}^{(i)}}\right\}. \tag{B.1}$$

As mentioned in Section 2.2, $P\left(o_\tau^{(i)}\middle|s_\tau^{(j)}, A^{(ij)}, C_{Aij}\right)$ and $P\left(s_\tau^{(i)}\middle|s_{\tau-1}^{(j)}, B^{(ij)}, C_{Bij}\right)$ are given by categorical distributions $P\left(o_\tau^{(i)}\middle|s_\tau^{(j)}, A^{(ij)}, C_{Aij}\right) = \mathrm{Cat}\left(\left(A^{(ij)\odot C_{Aij}^{\ddagger}}\right)s_\tau^{(j)}\right)$ and $P\left(s_\tau^{(i)}\middle|s_{\tau-1}^{(j)}, B^{(ij)}, C_{Bij}\right) = \mathrm{Cat}\left(\left(B^{(ij)\odot C_{Bij}^{\ddagger}}\right)s_{\tau-1}^{(j)}\right)$, respectively. As C_A takes the value of 1 or 0, the following equation holds:

$$A^{(ij)\odot C_{Aij}^{\ddagger}} = \begin{cases} A^{(ij)} & (C_{Aij} = 1) \\ \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} & (C_{Aij} = 0) \end{cases}, \tag{B.2}$$

which means that $C_{Aij} = 0$ reduces $A^{(ij)}$ from the model owing to the factorial property of the model. The same applies to B. The prior beliefs about $A^{(ij)}$, $B^{(ij)}$ and C_{Aij}, C_{Bij} follow the Dirichlet distributions $P\left(A^{(ij)}\right) = \mathrm{Dir}\left(a^{(ij)}\right)$, $P\left(B^{(ij)}\right) = \mathrm{Dir}\left(b^{(ij)}\right)$ and categorical distributions $P\left(C_{Aij}\right) = \mathrm{Cat}\left(c_{Aij}\right)$, $P\left(C_{Bij}\right) = \mathrm{Cat}\left(c_{Bij}\right)$, respectively. The initial states follow a categorical distribution: $P(s_1) = \mathrm{Cat}(D)$. The partition functions $Z_{C_A}^{(i)}$ and $Z_{C_B}^{(i)}$ serve as normalization factors to counterbalance the deviations of $\sum_{o_\tau^{(i)}=1,0}\prod_{j=1}^{N_s} P\left(o_\tau^{(i)}\middle|s_\tau^{(j)}, A^{(ij)}, C_{Aij}\right)$ and $\sum_{o_\tau^{(i)}=1,0}\prod_{j=1}^{N_s} P\left(s_\tau^{(i)}\middle|s_{\tau-1}^{(j)}, B^{(ij)}, C_{Bij}\right)$ from 1, which are expressed as

$$Z_{C_A}^{(i)} = \sum_{k=1}^{2}\prod_{j=1}^{N_s}\left(A_{k1}^{(ij)^{C_{Aij}}} \quad A_{k2}^{(ij)^{C_{Aij}}}\right)s_\tau^{(j)},\ Z_{C_B}^{(i)} = \sum_{k=1}^{2}\prod_{j=1}^{N_s}\left(B_{k1}^{(ij)^{C_{Bij}}} \quad B_{k2}^{(ij)^{C_{Bij}}}\right)s_{\tau-1}^{(j)}. \tag{B.3}$$

Appendix C. Detailed description of variational free energy

Here, we describe the explicit forms of posterior beliefs and variational free energy and derive belief update rules that minimize variational free energy. The explicit form of Eq. (10) (adopting the factorial structure) can be expressed as

$$Q(s_{1:t},\theta)=\prod_{i=1}^{N_o}\prod_{j=1}^{N_s}\{Q(A^{(i,j)})Q(C_{Aij})\}\cdot\prod_{i=1}^{N_s}\prod_{j=1}^{N_s}\{Q(B^{(i,j)})Q(C_{Bij})\}\cdot\prod_{\tau=1}^{t}\prod_{j=1}^{N_s}Q(s_\tau^{(j)}). \tag{C.1}$$

The marginal posterior beliefs about $A^{(i,j)}$, $B^{(i,j)}$, C_{Aij}, C_{Bij}, $s_\tau^{(j)}$ follow the Dirichlet distributions $Q(A^{(i,j)})=\mathrm{Dir}(\mathbf{a}^{(i,j)})$, $Q(B^{(i,j)})=\mathrm{Dir}(\mathbf{b}^{(i,j)})$ and categorical distributions $Q(C_{Aij})=\mathrm{Cat}((\mathbf{c}_{Aij},\overline{\mathbf{c}_{Aij}})^{\mathrm{T}})$, $Q(C_{Bij})=\mathrm{Cat}((\mathbf{c}_{Bij},\overline{\mathbf{c}_{Bij}})^{\mathrm{T}})$, $Q(s_\tau^{(j)})=\mathrm{Cat}(\mathbf{s}_\tau^{(j)})$, respectively.

Under the generative model defined in Eq. (9), or Eq. (B.1), the variational free energy is expressed as follows:

$$F(o_{1:t},\ Q(s_{1:t},\theta))=\sum_{\tau=1}^{t}\mathrm{E}_Q[-\ln P(o_\tau|s_\tau,A,\ C_A)-\ln P(s_\tau|s_{\tau-1},B,C_B)+\ln Q(s_\tau)]+\mathscr{D}_{\mathrm{KL}}[Q(\theta)\|\ P(\theta)]$$

$$=\sum_{\tau=1}^{t}\left\{\mathbf{s}_\tau\cdot\left(\ln\mathbf{s}_\tau-\mathbf{c}_A^{\ddagger}\odot\ln\mathbf{A}\cdot o_\tau-\mathbf{c}_B^{\ddagger}\ln\mathbf{B}\mathbf{s}_{\tau-1}\right)+E_Q[\ln Z_{C_A}]+E_Q[\ln Z_{C_B}]\right\}$$

$$+(\mathbf{a}-a)\cdot\ln\mathbf{A}-\ln\mathscr{B}(\mathbf{a})+(\mathbf{b}-b)\cdot\ln\mathbf{B}-\ln\mathscr{B}(\mathbf{b})$$

$$+\left(\frac{\mathbf{c}_A}{\overline{\mathbf{c}}_A}\right)\cdot\left(\ln\left(\frac{\mathbf{c}_A}{\overline{\mathbf{c}}_A}\right)-\ln\left(\frac{c_A}{\overline{c}_A}\right)\right)+\left(\frac{\mathbf{c}_B}{\overline{\mathbf{c}}_B}\right)\cdot\left(\ln\left(\frac{\mathbf{c}_B}{\overline{\mathbf{c}}_B}\right)-\ln\left(\frac{c_B}{\overline{c}_B}\right)\right). \tag{C.2}$$

Note that factorial structures (i.e., summations over N_s and N_o) are implicitly involved for simplification. In the above, $\mathscr{B}\left(\mathbf{a}_l^{(i,j)}\right):=\Gamma\left(\mathbf{a}_{1l}^{(i,j)}\right)\Gamma\left(\mathbf{a}_{2l}^{(i,j)}\right)/\Gamma\left(\mathbf{a}_{1l}^{(i,j)}+\mathbf{a}_{2l}^{(i,j)}\right)$ denotes the beta function characterized by the gamma function $\Gamma(\cdot)$.

Appendix D. Approximation of partition functions

Here, we derive the approximation of partition function: $\mathrm{E}_Q[\ln Z_{C_A}]\approx\ln 2\cdot\sum_{i=1}^{N_o}\left(1-\sum_{j=1}^{N_s}\mathbf{c}_{A_{ij}}\right)$. When $A^{(i,j)}$ is expressed as $A^{(i,j)}=\begin{pmatrix}0.5+\Delta_{ij1} & 0.5+\Delta_{ij2}\\ 0.5-\Delta_{ij1} & 0.5-\Delta_{ij2}\end{pmatrix}$, the following equation holds:

$$Z_{C_A}^{(i)}:=\prod_{j=1}^{N_s}A_{1\cdot}^{(i,j)^{C_{Aij}}}s_\tau^{(j)}+\prod_{j=1}^{N_s}A_{2\cdot}^{(i,j)^{C_{Aij}}}s_\tau^{(j)}$$

$$=\prod_{j=1}^{N_s}\left((0.5,\ 0.5)+\Delta_{ij\cdot}\right)^{C_{Aij}}s_\tau^{(j)}+\prod_{j=1}^{N_s}\left((0.5,\ 0.5)-\Delta_{ij\cdot}\right)^{C_{Aij}}s_\tau^{(j)}, \tag{D.1}$$

where $\Delta_{ij\cdot}:=\left(\Delta_{ij1},\ \Delta_{ij2}\right)$. Then we obtain the following approximation by applying a first order approximation to Δ.

$$Z_{C_A}^{(i)}\approx\prod_{j=1}^{N_s}(0.5,\ 0.5)^{C_{Aij}}s_\tau^{(j)}\left\{1+\frac{\Delta_{i1\cdot}^{C_{Ai1}}s_\tau^{(1)}}{\left((0.5,\ 0.5)^{C_{Ai1}}\right)s_\tau^{(1)}}+\cdots+\frac{\Delta_{iN_s\cdot}^{C_{AiN_s}}s_\tau^{(N_s)}}{\left((0.5,\ 0.5)^{C_{AiN_s}}\right)s_\tau^{(N_s)}}\right\}+\prod_{j=1}^{N_s}(0.5,\ 0.5)^{C_{Aij}}s_\tau^{(j)}\left\{1-\frac{\Delta_{i1\cdot}^{C_{Ai1}}s_\tau^{(1)}}{\left((0.5,\ 0.5)^{C_{Ai1}}\right)s_\tau^{(1)}}-\cdots\right.$$
$$\left.-\frac{\Delta_{iN_s\cdot}^{C_{AiN_s}}s_\tau^{(N_s)}}{\left((0.5,\ 0.5)^{C_{AiN_s}}\right)s_\tau^{(N_s)}}\right\}$$

$$=2\prod_{j=1}^{N_s}\left((0.5\ 0.5)^{C_{Aij}}\right)s_\tau^{(j)}=2\prod_{j=1}^{N_s}0.5^{C_{Aij}}. \tag{D.2}$$

The last equality holds because $\sum_{l=1}^{2}s_{tl}^{(j)}=1$ holds. With this approximation, $\mathrm{E}_Q\left[\ln Z_{C_A}^{(i)}\right]$ is calculated as follows:

$$\mathrm{E}_Q\left[\ln Z_{C_A}^{(i)}\right]\approx\ln 2+\mathrm{E}_{Q(C_{Aij})}\left[\sum_{j=1}^{N_s}C_{Aij}\ln 0.5\right]=\ln 2+\sum_{j=1}^{N_s}\mathbf{c}_{A_{ij}}\ln 0.5=\ln 2\cdot\left(1-\sum_{j=1}^{N_s}\mathbf{c}_{A_{ij}}\right). \tag{D.3}$$

Therefore, $\mathrm{E}_Q[\ln Z_{C_A}]=\mathrm{E}_Q\left[\sum_{i=1}^{N_o}\ln Z_{C_A}^{(i)}\right]\approx\ln 2\cdot\sum_{i=1}^{N_o}\left(1-\sum_{j=1}^{N_s}\mathbf{c}_{A_{ij}}\right)$. In the same manner, $\mathrm{E}_Q\left[\ln Z_{B^{c_B}s_{\tau-1}}\right]\approx\ln 2\cdot\sum_{i=1}^{N_s}\left(1-\sum_{j=1}^{N_s}\mathbf{c}_{B_{ij}}\right)$ is derived.

Appendix E. Detailed derivation of the update rules from variational free energy

Here, we derive Eqs. (13 − 17) by taking derivatives of variational free energy F in Eq. (8). When the $\mathscr{O}(\ln t)$ term is written down explicitly, Eq. (8) is expressed as Eq. (C.2). First, differentiating F with respect to $\mathbf{s}_\tau^{(n)}$ yields the following:

$$\frac{\partial F}{\partial\mathbf{s}_\tau^{(n)}}=\ln\mathbf{s}_\tau^{(n)}-\sum_{i=1}^{N_o}\mathbf{c}_{Ain}\ln\mathbf{A}^{(i,n)}\cdot o_\tau^{(i)}-\sum_{j=1}^{N_s}\mathbf{c}_{Bnj}\ln\mathbf{B}^{(n,j)}\mathbf{s}_{\tau-1}^{(j)}+1. \tag{E.1}$$

Thus, at the fixed point where $\partial F/\partial \mathbf{s}_\tau^{(n)} = 0$ holds, the following holds:

$$\ln \mathbf{s}_\tau^{(n)} = \sum_{i=1}^{N_o} \mathbf{c}_{Ain} \ln \mathbf{A}^{(i,n)} \cdot o_\tau^{(i)} + \sum_{j=1}^{N_s} \mathbf{c}_{Bnj} \ln \mathbf{B}^{(n,j)} \mathbf{s}_{\tau-1}^{(j)} - 1. \tag{E.2}$$

Considering that the sum of $\mathbf{s}_\tau^{(n)}$ need to be one, solving the fixed point of the implicit gradient descent $\partial F/\partial \mathbf{s}_\tau^{(n)} = 0$ yields Eq. (13).

Next, differentiating F with respect to $\mathbf{a}^{(i,j)}$ yields the following:

$$\frac{\partial F}{\partial \mathbf{a}^{(ij)}} = -\mathbf{c}_{Aij} \frac{\partial \ln \mathbf{A}^{(ij)}}{\partial \mathbf{a}^{(ij)}} \cdot \sum_{\tau=1}^{t} o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)} + \ln \mathbf{A}^{(ij)} + \left(\mathbf{a}^{(ij)} - a^{(ij)}\right) \cdot \frac{\partial \ln \mathbf{A}^{(ij)}}{\partial \mathbf{a}^{(ij)}} - \frac{\partial \ln \mathscr{B}\left(\mathbf{a}^{(ij)}\right)}{\partial \mathbf{a}^{(ij)}}. \tag{E.3}$$

Here, because $\frac{\partial \ln \mathscr{B}\left(\mathbf{a}^{(ij)}\right)}{\partial \mathbf{a}^{(ij)}} = \ln \mathbf{A}^{(ij)}$ holds, Eq. (E.3) can be transformed as follows:

$$\frac{\partial F}{\partial \mathbf{a}^{(ij)}} = \frac{\partial \ln \mathbf{A}^{(ij)}}{\partial \mathbf{a}^{(ij)}} \cdot \left(\mathbf{a}^{(ij)} - a^{(ij)} - \mathbf{c}_{Aij} \sum_{\tau=1}^{t} o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)} \right). \tag{E.4}$$

Therefore, $\partial F/\partial \mathbf{a}^{(ij)} = 0$ yields Eq. (14). In addition, $\partial F/\partial \mathbf{b}^{(ij)} = 0$ yields Eq. (15) in the same manner.

Finally, differentiating F with respect to $\mathbf{c}_{Aij}$ yields the following:

$$\frac{\partial F}{\partial \mathbf{c}_{Aij}} = -\sum_{\tau=1}^{t} \left(\ln \mathbf{A}^{(ij)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right) + \ln 2 \right) + \mathrm{sig}^{-1}\left(\mathbf{c}_{Aij}\right) - \mathrm{sig}^{-1}\left(c_{Aij}\right). \tag{E.5}$$

Note that here we use $\ln c - \ln \bar{c} = \mathrm{sig}^{-1}(c)$. Therefore, $\partial F/\partial \mathbf{c}_{Aij} = 0$ yields Eq. (16). In the same manner, $\partial F/\partial \mathbf{c}_{Bij} = 0$ yields Eq. (17).

Appendix F. Properties of connectivity parameters

Here, we show that $\mathbf{c}_{Aij}$ converges to 0 when $o_\tau^{(i)}$ is randomly determined and independent of all elements of s_τ (which is defined as the condition). Moreover, we consider the case where the likelihood mapping posterior $\mathbf{A}^{(ij)}$ is sufficiently flat (assumption 1) and the mean state posterior $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]$ is sufficiently flat (assumption 2) for analytical tractability.

The direction of the change in $\mathbf{c}_{Aij}$ is determined by the sign of the second term on the right-hand side of Eq. (16). Because $o_\tau^{(i)}$ changes randomly from the condition, i.e., $p\left(o_{\tau 1}^{(i)} = 1 | s_\tau\right) = p\left(o_{\tau 2}^{(i)} = 1 | s_\tau\right) = 1/2$, the expectation of $\ln \mathbf{A}^{(ij)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right) + \ln 2$ over observations (i.e., the negative cross entropy term) is calculated as follows:

$$\begin{aligned}
&\mathrm{E}_{P(o_\tau)}\left[\ln \mathbf{A}^{(ij)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right) + \ln 2\right] \\
&= \frac{1}{2}\left(\ln \mathbf{A}^{(ij)} \cdot \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \mathbf{s}'^{(j)}_\tau \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \right\} + \ln 2 \right) + \frac{1}{2}\left(\ln \mathbf{A}^{(ij)} \cdot \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \mathbf{s}'^{(j)}_\tau \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \right\} + \ln 2 \right),
\end{aligned} \tag{F.1}$$

where $\mathbf{s}'^{(j)}_\tau\left(o_\tau^{(i)}\right) := \mathrm{E}_{P\left(o_\tau^{(1)}, \ldots, o_\tau^{(i-1)}, o_\tau^{(i+1)}, \ldots, o_\tau^{(N_o)}\right)}\left[\mathbf{s}_\tau^{(j)}\right]$ denotes the expected state posterior inferred after observing only $o_\tau^{(i)}$. This is a random variable that depends on $o_\tau^{(i)}$. Eq. (F.1) is further transformed as follows:

$$\begin{aligned}
&\mathrm{E}_{P(o_\tau)}\left[\ln \mathbf{A}^{(ij)} \cdot \left(o_\tau^{(i)} \otimes \mathbf{s}_\tau^{(j)}\right) + \ln 2\right] \\
&= \frac{1}{2} \mathbf{s}'^{(j)}_\tau \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \cdot \left(\ln \mathbf{A}^{(ij)}_{1\cdot} + \ln 2 \right) + \frac{1}{2} \mathbf{s}'^{(j)}_\tau \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \cdot \left(\ln \mathbf{A}^{(ij)}_{2\cdot} + \ln 2 \right).
\end{aligned} \tag{F.2}$$

Then, the following approximation holds when $\mathbf{a}^{(ij)}$ is sufficiently large: $\ln \mathbf{A}^{(ij)} := \mathrm{E}_{Q\left(A^{(ij)}\right)}\left[\ln A^{(ij)}\right] \approx \ln \mathrm{E}_{Q\left(A^{(ij)}\right)}\left[A^{(ij)}\right]$. It is owing to the properties of the digamma functions (Isomura & Friston, 2020). Subsequently, we define the small deviations Δ_{ij1}, Δ_{ij2} as the values that satisfy the following (i.e., assumption 1): $\ln \mathrm{E}_{Q\left(A^{(ij)}\right)}\left[A^{(ij)}\right] = \ln \begin{pmatrix} 0.5 + \Delta_{ij1} & 0.5 + \Delta_{ij2} \\ 0.5 - \Delta_{ij1} & 0.5 - \Delta_{ij2} \end{pmatrix}$. With these, the Taylor expansion of $\ln \mathbf{A}^{(ij)}$ at $\frac{1}{2}$ yields the following equation:

$$\ln \mathbf{A}^{(ij)} = -\ln 2 + 2 \begin{pmatrix} \Delta_{ij1} & \Delta_{ij2} \\ -\Delta_{ij1} & -\Delta_{ij2} \end{pmatrix} - 2 \begin{pmatrix} \Delta_{ij1}^2 & \Delta_{ij2}^2 \\ \Delta_{ij1}^2 & \Delta_{ij2}^2 \end{pmatrix} + \mathscr{O}\left(\Delta_{ij}^{\odot 3}\right). \tag{F.3}$$

Furthermore, we consider new variable $S := \mathrm{E}_{P\left(o_\tau^{(1)}, \ldots, o_\tau^{(i-1)}, o_\tau^{(i+1)}, \ldots, o_\tau^{(N_o)}\right)}\left[\sigma\left(\sum_{i' \neq i} \mathbf{c}_{Ai'j} \ln \mathbf{A}^{(i'j)} \cdot o_\tau^{(i')}\right)\right]$. Combining Eq. (13) with S and Eq. (F.3), the Taylor expansion of $\mathbf{s}'^{(j)}_\tau\left(o_\tau^{(i)}\right)$ can be expressed:

$$\mathbf{s}'^{(j)}_\tau\left(o_\tau^{(i)}\right) \approx S + 2\mathbf{c}_{Aij}\left(o_{\tau 1}^{(i)} - o_{\tau 2}^{(i)}\right) S \odot (1 - S) \odot \Delta_{ij} + \mathscr{O}\left(\Delta_{ij}^{\odot 2}\right). \tag{F.4}$$

Here, we used the Taylor expansion of sigmoid function: $\sigma(a + x) = \sigma(a) + \sigma(a)(1 - \sigma(a))x + \mathscr{O}(x^2)$, and the property of sigmoid function: $\sigma\left(x + \beta \overrightarrow{1}\right) = \sigma(x)$ where β is an arbitrary scalar.

With Eq. (F.3) and Eq. (F.4), the right-hand side of Eq. (F.2) is further calculated as follows:

$$\mathrm{E}_{P(o_\tau)}\left[\ln\mathbf{A}^{(i,j)}\cdot\left(o_\tau^{(i)}\otimes\mathbf{s}_\tau^{(j)}\right)+\ln 2\right]=4S\odot\left\{-1+2\mathbf{c}_{Aij}(1-S)\right\}\cdot\Delta_{ij}^{\odot 2} \tag{F.5}$$

Here, the terms with $\mathscr{O}\left(\Delta_{ij}^{3}\right)$ or higher order were ignored owing to Δ_{ij1} and Δ_{ij2} that are sufficiently close to 0 (i.e., assumption 1).

From Eq. (F.4), the mean state posterior $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]$ is computed as $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]=\mathrm{E}_{P\left(o_\tau^{(i)}\right)}\left[\mathbf{s}'^{(j)}_\tau\left(o_\tau^{(i)}\right)\right]=S+\mathscr{O}\left(\Delta_{ij}^{\odot 2}\right)$. Because of assumption 2, $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]=\begin{pmatrix}0.5+\epsilon\\0.5-\epsilon\end{pmatrix}$ holds with small ϵ. Therefore, $S=\begin{pmatrix}0.5+\epsilon\\0.5-\epsilon\end{pmatrix}+\mathscr{O}\left(\Delta_{ij}^{\odot 2}\right)$ holds. Substituting this into Eq. (F.5) yields the following:

$$\mathrm{E}_{P(o_\tau)}\left[\ln\mathbf{A}^{(i,j)}\cdot\left(o_\tau^{(i)}\otimes\mathbf{s}_\tau^{(j)}\right)+\ln 2\right]=2\left\{-\begin{pmatrix}1+2\epsilon\\1-2\epsilon\end{pmatrix}+\mathbf{c}_{Aij}\begin{pmatrix}1-4\epsilon^2\\1-4\epsilon^2\end{pmatrix}\right\}\cdot\Delta_{ij}^{\odot 2}$$

$$=2\left(\Delta_{ij1}{}^2+\Delta_{ij2}{}^2\right)\left(\mathbf{c}_{Aij}-1-2\epsilon\cdot\frac{\Delta_{ij1}{}^2-\Delta_{ij2}{}^2}{\Delta_{ij1}{}^2+\Delta_{ij2}{}^2}\right). \tag{F.6}$$

Because ϵ is small, terms with ϵ^2 were ignorable in the last transformation. This means that equation (F.6) is negative when $\mathbf{c}_{Aij}<1+2\epsilon\cdot\frac{\Delta_{ij1}{}^2-\Delta_{ij2}{}^2}{\Delta_{ij1}{}^2+\Delta_{ij2}{}^2}$ holds. This is satisfied when assumptions 1 and 2 (which assure that Δ_{ij1}, Δ_{ij2} and ϵ are small) hold and when the prior of connectivity (i.e., initial value of $\mathbf{c}_{Aij}$) is not sufficiently close to 1.

Finally, $\sum_{\tau=1}^{t}\left(\ln\mathbf{A}^{(i,j)}\cdot\left(o_\tau^{(i)}\otimes\mathbf{s}_\tau^{(j)}\right)+\ln 2\right)$ has a $|\Delta_{ij}|\sqrt{t}$-order standard deviation according to the central limit theorem when $o_\tau^{(i)}\otimes\mathbf{s}_\tau^{(j)}$ is read as a stochastic variable. Taken together, $\sum_{\tau=1}^{t}\left(\ln\mathbf{A}^{(i,j)}\cdot\left(o_\tau^{(i)}\otimes\mathbf{s}_\tau^{(j)}\right)+\ln 2\right)$ is a $|\Delta_{ij}|^2 t$-order negative value with a $|\Delta_{ij}|\sqrt{t}$-order standard deviation, which decreases monotonically with time when t is sufficiently large. Therefore, the direction of change in $\mathbf{c}_{Aij}$ is negative when $o_\tau^{(i)}$ is randomly determined and independent of all elements of s_τ, and when $\mathbf{A}^{(i,j)}$ and $\mathrm{E}_{P(o_\tau)}\left[\mathbf{s}_\tau^{(j)}\right]$ are sufficiently flat. Hence, $\mathbf{c}_{Aij}$ converges to 0 after sufficient sessions.

Appendix G. Numerical simulation 1: Blind source separation

The agents received 1024 (i.e., 32 × 32 pixels) observations that were generated using the generative model defined in Eq. (3). The model comprised two states, $s^{(1)}$ and $s^{(2)}$, which generated the observations. Both states randomly took values of 0 or 1 with a probability of 0.5 each (as depicted by the flat matrix B with all elements of 0.5). When $s^{(1)}$ or $s^{(2)}$ was ON, the shape of a circle or rectangle was displayed on the screen as sensory input for the agent (Fig. 2a). This was implemented using the following matrix A:

$$A^{(i,1)}=\begin{cases}\begin{pmatrix}0.5&0.5\\0.5&0.5\end{pmatrix}\ (i\notin\text{the points on a circle})\\ \begin{pmatrix}0.6&0.4\\0.4&0.6\end{pmatrix}\ (i\in\text{the points on a circle})\end{cases}, \tag{G.1}$$

$$A^{(i,2)}=\begin{cases}\begin{pmatrix}0.5&0.5\\0.5&0.5\end{pmatrix}\ (i\notin\text{the points on a rectangle})\\ \begin{pmatrix}0.6&0.4\\0.4&0.6\end{pmatrix}\ (i\in\text{the points on a rectangle})\end{cases}. \tag{G.2}$$

The agent optimized $\mathbf{s}$, $\mathbf{a}$, and $\mathbf{c}_\mathbf{A}$ (i.e., $\mathbf{b}$ and $\mathbf{c}_\mathbf{B}$ were omitted from this simulation). The prior beliefs about A and C_A characterized the inference of the agents. The elements of prior belief $a^{(i,j)}$ with the i-index denoting a point on the shape is expressed as follows: $a^{(i,j)}=\begin{pmatrix}25.5&24.5\\24.5&25.5\end{pmatrix}$. The other elements $a^{(i,j)}$ were randomly sampled from a uniform distribution with the center of 25 and interval of 1. All elements of prior belief about C_A were sampled from a uniform distribution with the center of 0.9 and interval of 0.1. Further details are provided in the source code, which can be downloaded from the following link: https://github.com/UTTazawa/ProjectBSyMP.

The full model and BSyMP agents optimized their posterior beliefs: they optimized $\mathbf{s}_\tau$ at every time point and optimized $\mathbf{a}$ and $\mathbf{c}_\mathbf{A}$ at every 10 time points. Following optimization over 10,000 time points, BMR was applied to the learned full model (Friston et al., 2019). BMR computed the change in free energy $\Delta F^{(i,j)}$ when each $a^{(i,j)}$ was varied. A prior for reduction $\tilde{a}^{(i,j)}$ was set to a uniform matrix with large positive values. The element of the posterior $\mathbf{a}^{(i,j)}$ was reduced if the change in the approximate free energy $\Delta F^{(i,j)}:=\ln\mathscr{B}\left(\tilde{a}^{(i,j)}\right)-\ln\mathscr{B}\left(\tilde{\mathbf{a}}^{(i,j)}\right)-\ln\mathscr{B}\left(a^{(i,j)}\right)+\ln\mathscr{B}\left(\mathbf{a}^{(i,j)}\right)$ was negative and retained otherwise. Further details regarding BMR can be found in Friston et al., 2019.

After performing BMR, the variational free energy was calculated by following Eq. (5) at the end of the training sessions, however, the complexity of the connectivity parameters was excluded for a consistent evaluation among the three methods. Subsequently, the normalized mapping posteriors for all agents were calculated. The normalized mapping posterior $\mathbf{A}$ for the full model and BMR and $\mathbf{A}^{\odot\mathbf{c}_A^\dagger}$ for BSyMP are expressed as follows:

$$\mathbf{A}_{kl}^{(i,j)}=\frac{\mathbf{a}_{kl}^{(i,j)}}{\mathbf{a}_{1l}^{(i,j)}+\mathbf{a}_{2l}^{(i,j)}},\ \mathbf{A}^{\odot\mathbf{c}_A^\dagger}{}_{kl}^{(i,j)}=\frac{\left(\mathbf{a}_{kl}^{(i,j)}\right)^{\mathbf{c}_{Aij}}}{\left(\mathbf{a}_{1l}^{(i,j)}\right)^{\mathbf{c}_{Aij}}+\left(\mathbf{a}_{2l}^{(i,j)}\right)^{\mathbf{c}_{Aij}}}. \tag{G.3}$$

Finally, 1000 time-point-length data points were created using the same generative process to test the performance of the agents.

Appendix H. Performance of online version of BMR on the BSS task

As noted in Section 3.1, the underperformance of BMR in the BSS task can be attributed to the post hoc model reduction. Here we evaluated the online version of BMR (Friston et al., 2017). The online BMR agent performed BMR on each session and continued inference and learning with the reduced posterior. This indicates that once a parameter was reduced in a session, it remained reduced in the subsequent sessions. Thus, to reduce only the parameters with high confidences for the reduction, the element $\mathbf{a}^{(i,j)}$ was reduced when $\Delta F^{(i,j)} < -3$ held.

The performance of the online BMR agent was almost the same level as that of the BMR agent (Fig. H.1). This is arguably because the careful reduction led to small effect on the mapping posterior at the early stage of learning, resulting in the failure in modifying the learning trajectories from those of the full model. We also evaluated online BMR that reduced $\mathbf{a}^{(i,j)}$ when $\Delta F^{(i,j)} < 0$ held—the same threshold for the post hoc version—almost all the parameters were reduced in this case, resulting in the performance close to the chance level (data not shown). In summary, online BMR as well as conventional BMR failed to perform the BSS task because of the heuristic in the threshold for $\Delta F^{(i,j)}$.

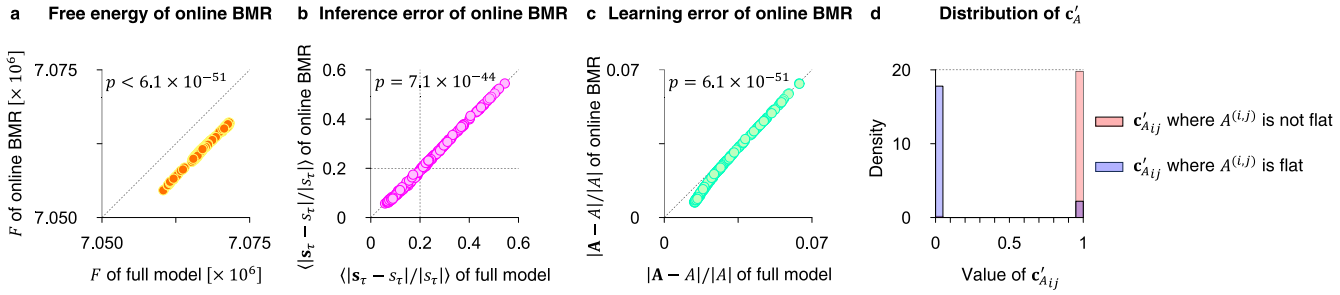


Fig. H.1. Performance of online version of BMR on the BSS task. (**a**) A comparison of the free energy between online BMR and the full model. (**b**) A comparison of the state inference error between online BMR and the full model. (**c**) A comparison of the parameter learning error between online BMR and the full model. (**d**) The distribution of the online BMR's counterpart, $\mathbf{c}'_A$, to the connectivity parameter in BSyMP, $\mathbf{c}_A$, at the end of the training sessions.

Appendix I. Analysis of the reduction by BMR

Here we analyze the mechanism of BMR reducing an element of A matrix. BMR reduces the element if the change in free energy $\Delta F^{(i,j)} := F_{\mathrm{red}(i,j)} - F_{\mathrm{full}}$ is negative, where F_{full} and $F_{\mathrm{red}(i,j)}$ are free energies of full model and that with the reduced (i,j) connectivity, respectively. Under the POMDP generative model, this is computed as $\Delta F^{(i,j)} = \ln\mathscr{B}(\widetilde{a}^{(i,j)}) - \ln\mathscr{B}(\widetilde{\mathbf{a}}^{(i,j)}) - \ln\mathscr{B}(a^{(i,j)}) + \ln\mathscr{B}(\mathbf{a}^{(i,j)}) = \sum_{l=1}^{2}\left\{\ln\mathscr{B}(\widetilde{a}_{\cdot l}^{(i,j)}) - \ln\mathscr{B}(\widetilde{\mathbf{a}}_{\cdot l}^{(i,j)}) - \ln\mathscr{B}\left(a_{\cdot l}^{(i,j)}\right) + \ln\mathscr{B}\left(\mathbf{a}_{\cdot l}^{(i,j)}\right)\right\}$. We analyze the behavior of $\Delta F_{\cdot l}^{(i,j)} := \ln\mathscr{B}(\widetilde{a}_{\cdot l}^{(i,j)}) - \ln\mathscr{B}(\widetilde{\mathbf{a}}_{\cdot l}^{(i,j)}) - \ln\mathscr{B}\left(a_{\cdot l}^{(i,j)}\right) + \ln\mathscr{B}\left(\mathbf{a}_{\cdot l}^{(i,j)}\right)$ under the following conditions:

$$\begin{aligned} &a_{\cdot l}^{(i,j)} = \begin{pmatrix} a \\ a \end{pmatrix},\ \mathbf{a}_{\cdot l}^{(i,j)} = m\begin{pmatrix} a+d \\ a-d \end{pmatrix},\ \widetilde{a}_{\cdot l}^{(i,j)} = \begin{pmatrix} \widetilde{a} \\ \widetilde{a} \end{pmatrix}, \\ &1 \le m,\ -\frac{m-1}{m}a \le d \le \frac{m-1}{m}a,\ \widetilde{a} \gg ma. \end{aligned} \tag{I.1}$$

These conditions enable us to handle $\mathbf{a}_{\cdot l}^{(i,j)}$ as a function of m and d, and interpret the meaning of the posterior easily. Coefficient m increases as the learning proceeds, and bias d denotes how $\mathbf{a}_{\cdot l}^{(i,j)}$ is far from a flat posterior. The ranges of m and d are determined to assure $a_{kl}^{(i,j)} \le \mathbf{a}_{kl}^{(i,j)}$. With these notations, $\Delta F_{\cdot l}^{(i,j)}$ is calculated as follows:

$$\Delta F_{\cdot l}^{(i,j)} \approx m\{(a+d)\ln(a+d) + (a-d)\ln(a-d) - 2a\ln a\} + \frac{1}{2}\left(\ln\frac{a}{a+d} + \ln\frac{a}{a-d} - \ln m\right). \tag{I.2}$$

This calculation was based on Stirling's approximation: $\ln\Gamma(x) \approx x\ln x - x - \frac{1}{2}\ln x$. Taking derivative of $\Delta F_{\cdot l}^{(i,j)}$ regarding d tells us that $\Delta F_{\cdot l}^{(i,j)}$ is a convex function of d and $\underset{d}{\mathrm{argmin}}\ \Delta F_{\cdot l}^{(i,j)} = 0$ gives the minimum value in the range of Eq. (I.1). Thus, $\min_d \Delta F_{\cdot l}^{(i,j)} = -\frac{1}{2}\ln m (\le 0)$. When d is close to 0, the following approximation holds:

$$\Delta F_{\cdot l}^{(i,j)} \approx \Delta F_{\cdot l}^{(i,j)}\Big|_{d=0} + \left(\frac{\partial \Delta F_{\cdot l}^{(i,j)}}{\partial d}\Big|_{d=0}\right)d + \left(\frac{\partial^2 \Delta F_{\cdot l}^{(i,j)}}{\partial d^2}\Big|_{d=0}\right)d^2 = -\frac{1}{2}\ln m + \frac{2(ma+1)}{a^2}d^2. \tag{I.3}$$

Therefore, the solution of $\Delta F_{\cdot l}^{(i,j)} = 0$ is approximated as $d = \pm\frac{2}{a}\sqrt{\frac{\ln m}{ma+1}}$, indicating that the larger m becomes, the narrower the range of d that holds $\Delta F_{\cdot l}^{(i,j)} < 0$ becomes. In summary, the range of BMR reducing sub-matrix to flat becomes narrower monotonically as the learning proceeds. This behavior is equivalent to the weight-based rule that prunes synapses with small weights in neural networks.

Appendix J. Correspondence between BSyMP and BMR

Here we show the correspondence between BSyMP and the previously developed Bayesian model reduction (BMR) method. BMR compares the amount of free energy between the learned full model and reduced model with different priors. For the factorial PODMP model considered in this work, the reduction of an element of A matrix is achieved by replacing its flat prior $a^{(i,j)} = \begin{pmatrix} \alpha & \alpha \\ \alpha & \alpha \end{pmatrix}$ with strong flat prior $\widetilde{a}^{(i,j)} = \begin{pmatrix} \widetilde{\alpha} & \widetilde{\alpha} \\ \widetilde{\alpha} & \widetilde{\alpha} \end{pmatrix}$, where $0 < \alpha \ll \widetilde{\alpha}$. The element of the reduced posterior $\widetilde{\mathbf{a}}^{(i,j)}$ takes the form of $\widetilde{\mathbf{a}}^{(i,j)} = \widetilde{a}^{(i,j)} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau}^{(i)} \otimes \mathbf{s}_{\tau}^{(j)}$. Thus, the normalized reduced posterior $\widetilde{\mathbf{A}}_{kl}^{(i,j)}$ is provided as follows:

$$\widetilde{\mathbf{A}}_{kl}^{(i,j)} := \frac{\widetilde{\mathbf{a}}_{kl}^{(i,j)}}{\widetilde{\mathbf{a}}_{1l}^{(i,j)} + \widetilde{\mathbf{a}}_{2l}^{(i,j)}} = \frac{\widetilde{\alpha} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}}{\widetilde{\alpha} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 1}^{(i)} \mathbf{s}_{\tau l}^{(j)} + \widetilde{\alpha} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 2}^{(i)} \mathbf{s}_{\tau l}^{(j)}} = \frac{\widetilde{\alpha} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}}{2\widetilde{\alpha} + \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}}. \tag{J.1}$$

Conversely, the posterior belief obtained by BSyMP ($\mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl}$) takes the following form:

$$\mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl} := \frac{\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)^{\mathbf{c}_{Aij}}}{\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 1}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)^{\mathbf{c}_{Aij}} + \left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 2}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)^{\mathbf{c}_{Aij}}}. \tag{J.2}$$

Because we consider the case with a flat prior $a^{(i,j)} = \begin{pmatrix} \alpha & \alpha \\ \alpha & \alpha \end{pmatrix}$, $\mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl}$ can be transformed as follows:

$$\mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl} = \frac{\left(1 + \frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right)^{\mathbf{c}_{Aij}}}{\left(1 + \frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 1}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right)^{\mathbf{c}_{Aij}} + \left(1 + \frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 2}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right)^{\mathbf{c}_{Aij}}}. \tag{J.3}$$

As the Taylor expansion of $(1+x)^c$ at $x = 0$ yields $(1+x)^c = 1 + cx + \mathcal{O}(x^2) \approx 1 + cx$ for small x, we obtain the following approximation:

$$\begin{aligned} \mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl} &\approx \frac{1 + \left(\frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right) \mathbf{c}_{Aij}}{1 + \left(\frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 1}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right) \mathbf{c}_{Aij} + 1 + \left(\frac{2\left(\alpha + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau 2}^{(i)} \mathbf{s}_{\tau l}^{(j)}\right)}{\sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}} - 1\right) \mathbf{c}_{Aij}} \\ &= \frac{\alpha + \frac{1}{2}\left(\frac{1}{\mathbf{c}_{Aij}} - 1\right) \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)} + \sum_{\tau=1}^{t} \boldsymbol{o}_{\tau k}^{(i)} \mathbf{s}_{\tau l}^{(j)}}{2\left\{\alpha + \frac{1}{2}\left(\frac{1}{\mathbf{c}_{Aij}} - 1\right) \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}\right\} + \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}}. \end{aligned} \tag{J.4}$$

Therefore, Eq. (J.4) is homologous to Eq. (J.1) when setting $\widetilde{\alpha} = \alpha + \frac{1}{2}\left(\frac{1}{\mathbf{c}_{Aij}} - 1\right) \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}$, indicating the formal correspondence between BSyMP and BMR. This analysis further highlights an important contrast between BSyMP and BMR. BMR is a binary reduction method that reduces an element if the reduction decreases free energy since $\widetilde{\alpha}$ is heuristically set as a large value a priori. Conversely, BSyMP is a continuous reduction method that reduces an element depending on the value of $\alpha + \frac{1}{2}\left(\frac{1}{\mathbf{c}_{Aij}} - 1\right) \sum_{\tau=1}^{t} \mathbf{s}_{\tau l}^{(j)}$, which provides the Bayes-optimal solution for $\mathbf{A}^{\mathbf{c}_A\,(i,j)}_{kl}$.

Appendix K. Numerical simulation 2: Rule learning

The agents received 3072 (i.e., $32 \times 32 \times 3$ pixels) observations generated using the generative model defined in Eq. (3). The model comprised 12 states, each of which exhibited a certain shape on one of the three screens. Matrix A was set similarly to that in Simulation 1, but the fidelity of the observations was determined as follows:

$$A^{(i,j)} = \begin{cases} \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} & (i \notin \text{the points on a shape}) \\ \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} & (i \in \text{the points on a shape}) \end{cases}. \tag{K.1}$$

The state change followed the transition rule characterized by matrix B. All 2×2 submatrices were symmetric. Therefore, $B_{11}^{(i,j)}$ was sufficient to characterize $B^{(i,j)}$, as follows:

$$B^{(i,j)} = \begin{pmatrix} B_{11}^{(i,j)} & 1 - B_{11}^{(i,j)} \\ 1 - B_{11}^{(i,j)} & B_{11}^{(i,j)} \end{pmatrix}. \tag{K.2}$$

We prepared three transition rules by three matrices B (i.e., three sets of sub-matrices $B^{(i,j)}$). The transition rules are visualized in Fig. K.1. The initial states were determined randomly, as depicted by a flat block vector D with all elements of 0.5.

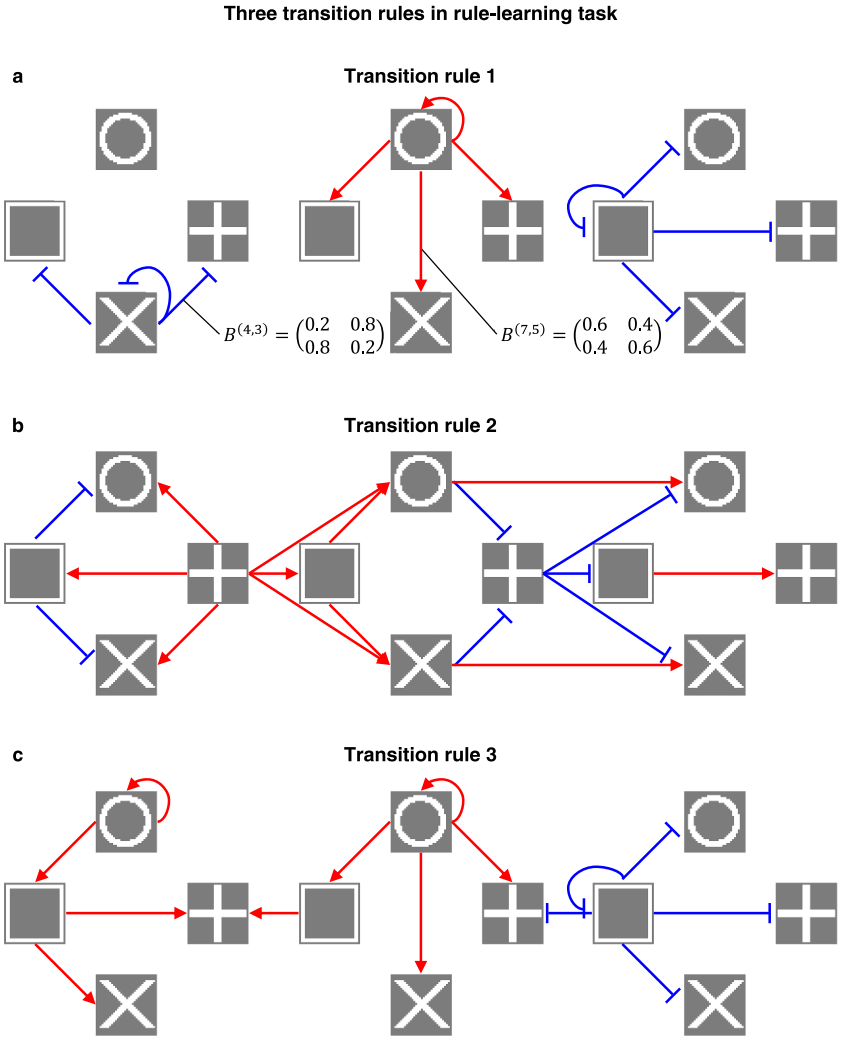


Fig. K.1. Three transition rules of rule-learning task. (**a**) The same transition rule displayed in Fig. 4a. Red arrows from one state $s^{(j)}$ to another state $s^{(i)}$ indicate that $B_{11}^{(i,j)}$ is larger than 0.5 (i.e., positive connection), and blue T-shaped lines indicate that $B_{11}^{(i,j)}$ is smaller than 0.5 (i.e., negative connection). (**b, c**) The other transition rules.

The agent employed the prior beliefs about A, B and C_A, C_B. The elements of prior belief $a^{(i,j)}$ were set as follows:

$$a^{(i,j)} = \begin{cases} \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} \cdot 10^5 \ (i \notin \text{the points on a shape}) \\ \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} \cdot 10^5 \ (i \in \text{the points on a shape}) \end{cases}. \tag{K.3}$$

All elements of the prior belief $b^{(i,j)}$ were sampled from a uniform distribution with the mean of 50 and interval of 2. All elements of the prior beliefs c_A and c_B were sampled from a uniform distribution with the mean of 0.9 and interval of 0.1. Further details are provided in the source code, which can be downloaded from the following link: https://github.com/UTTazawa/ProjectBSyMP.

The full model and BSyMP agents optimized their posterior beliefs: they optimized $\mathbf{s}_\tau$ at every time point and optimized $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}_A$, and $\mathbf{c}_B$ at every 100 time points. Following optimization over 10,000 time points, BMR was applied to the learned full model in the same manner as in Simulation 1.

After performing BMR, the variational free energy and the normalized transition rule posteriors $\mathbf{B}$ and $\mathbf{B}^{\odot \mathbf{c}_B^{\dagger}}$ were calculated in the same manner as in Simulation 1. Finally, 1000 time-point-length data points were created using the same generative process to test the prediction performance of the agents. The predicted state of the i th source was defined as $\mathbf{s}_{\tau|\tau-1}^{(i)} := \sigma\left(\sum_{j=1}^{N_s} \mathbf{c}_{Bij} \ln \mathbf{B}^{(i,j)} \mathbf{s}_{\tau-1}^{(j)}\right)$ for BSyMP and $\mathbf{s}_{\tau|\tau-1}^{(i)} := \sigma\left(\sum_{j=1}^{N_s} \ln \mathbf{B}^{(i,j)} \mathbf{s}_{\tau-1}^{(j)}\right)$ for the full model and BMR. The actual probability of the i th source state was defined as follows: $s_{\tau|\tau-1}^{(i)} = \sigma\left(\sum_{j=1}^{N_s} \ln B^{(i,j)} s_{\tau-1}^{(j)}\right)$.

Data availability

The code employed in this study is openly accessible at https://github.com/UTTazawa/ProjectBSyMP.

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